

Long-Term Memory Access and Its Impact on Multi-Session Performance in Small Language Models

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RECOMMENDATION SHEET

This is to certify that I have supervised the preparation of and read the thesis proposal prepared by A. Alon, John Kenneth, B. Gonzales, Rexter, C. Mariano, Erika entitled **Long-Term Memory Access and Its Impact on Multi-Session Performance in Small Language Models** and that the said paper has been submitted for final examination by the Oral Examination Committee.

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Chapter 1:

INTRODUCTION

This chapter establishes the research motivation by contextualizing the increasing prevalence and utility of Large Language Models (LLMs). Although LLMs are currently deployed across diverse applications ranging from retrieval-augmented generation (RAG) and chain-of-thought reasoning to complex agentic workflows their widespread adoption is frequently constrained by significant computational requirements. Consequently, the focus has shifted toward Small Language Models (SLMs) as efficient, hardware-accessible alternatives. However, a critical limitation persists regarding the capacity for sustained contextual history within these models. This study addresses the necessity for memory-augmented architectures capable of maintaining interaction history without exceeding the constraints of modest hardware. Furthermore, this chapter delineates the research gap, articulates the primary research objectives, and establishes the significance of the study, providing a theoretical foundation for examining how lightweight memory designs can enhance the continuity, reliability, and overall performance of SLM-based agents.

1.1 Context of the Study

Language Models are increasingly deployed in real-world applications, including education, healthcare, industry, and government services [1][2]. In educational platforms, conversational agents deliver adaptive tutoring and personalized feedback, while healthcare assistants summarize patient histories and suggest possible diagnoses. Across these domains, Language Models are expected to maintain contextual continuity, retain relevant prior information, and adapt responses based

on accumulated experience [2][3].

Most of these models are typically powered by large language models (LLMs), which exhibit strong reasoning capabilities and high-quality natural language understanding. However, these models often operate in a stateless manner, meaning each input prompt is processed independently without persistent storage of prior interactions. Stateless operation implies that previous user context, preferences, and task progress are not inherently retained unless explicitly reintroduced into the input [2][3][4]. This limitation creates significant challenges for long-running tasks that require continuity, personalization, and session-aware reasoning [4][5].

Packer et al. [5] demonstrated the efficacy of virtual context management through MemGPT, a system drawing inspiration from hierarchical memory systems in traditional operating systems. To address the severe handicap that limited context windows impose on modern Large Language Models (LLMs), MemGPT was specifically evaluated in the domain of document analysis. This evaluation proved that by intelligently managing data movement between fast and slow memory tiers, MemGPT can successfully analyze large documents that far exceed the underlying LLM’s fixed context constraints. Consequently, this establishes how structured memory architectures can artificially extend a model’s context window, acting as a robust retrieval mechanism for processing extensive textual data rather than functioning solely as an autonomous conversational agent.

Wu et al. [6] extended previous research and identified that systems such as MemGPT often lack advanced memory organization and adaptability across diverse tasks. This finding has motivated the development of more flexible, agentic structures that do not rely on fixed operational patterns. The concept of an “agentic memory system” is introduced, which enables agents to dynamically organize and evolve their memory structures, moving beyond static storage operations.

Moreover, Gutiérrez et al. [7] proposed a retrieval framework inspired by the hippocampal memory indexing theory of human Long Term Memory (LTM) to address the inability of standard Retrieval-Augmented Generation (RAG) sys-

tems to integrate knowledge across disjointed passages. This approach allows for single-step multi-hop reasoning by identifying paths between seemingly unrelated concepts, significantly outperforming iterative retrieval methods in efficiency.

The studies mentioned, while improving efficiency within their specific contexts, primarily rely on LLMs. Although LLMs possess considerable capabilities, their implementation requires substantial computational resources, significant memory usage, and continuous access to high-performance hardware [8][9]. These requirements limit their feasibility in environments with constrained resources, especially given that the computational complexity of the self-attention mechanism grows quadratically with sequence length [10]. In various practical scenarios, particularly in developing economies or installations requiring stringent privacy, using cloud-based execution for large models is often neither financially feasible nor operationally preferred [9].

Small Language Models (SLMs) have emerged as practical alternatives for these environments. SLMs typically contain significantly fewer parameters than LLMs and are optimized for efficient inference, reduced energy consumption, and local deployment [9][11]. Techniques such as knowledge distillation, parameter-efficient fine-tuning, and optimized attention mechanisms enable SLMs to achieve competitive performance on domain-specific tasks while remaining lightweight enough for modest hardware [9][12]. However, the reduced parameter capacity and narrower context windows of SLMs constrain their ability to internally store and recall extensive contextual knowledge. Unlike LLMs, which leverage broader reasoning capacity to filter and integrate retrieved information, SLMs operate under stricter computational and representational limitations [9][11].

One promising paradigm for addressing this limitation is Retrieval-Augmented Generation (RAG), introduced by Lewis et al. [13] as a framework that augments a language model’s parametric knowledge with dynamically retrieved external information at inference time. Rather than relying solely on knowledge encoded in model weights during training, RAG systems maintain a separate external knowl-

edge store and, upon receiving a query, retrieve the most relevant documents from that store before passing them alongside the original query into the model’s context window for generation. This retrieve-then-generate pipeline effectively decouples knowledge storage from model parameters, allowing the model to ground its responses in retrieved evidence rather than in compressed parametric representations alone. For SLMs operating under constrained context windows and limited reasoning bandwidth, RAG is particularly significant: it externalizes the memory burden that would otherwise overwhelm the model’s internal representations, enabling contextually accurate responses without requiring the parameter scale of frontier LLMs [8].

1.2 The Problem, Gap, or Opportunity

Existing LTM frameworks are primarily designed and evaluated for LLMs [1][2][3]. These frameworks often depend on computationally expensive retrieval processes, high-dimensional embeddings, and multi-step reasoning to identify relevant memory entries [2][5]. When applied to SLMs, such approaches may introduce several limitations: retrieval noise can overwhelm the model’s limited reasoning bandwidth, excessive memory context may exceed the available token window, and additional retrieval steps can increase latency beyond acceptable thresholds for real-time deployment [5][7][8]. Consequently, there is a need to systematically investigate how different LTM architectures behave when integrated with SLM operating under resource constraints [8][9].

To address this gap, this study investigates the integration of LTM modules within SLMs using two distinct memory designs: a retrieval-based store for semantic history and a cache-like mechanism for persistent contextual summaries [14][12][6]. Unlike prior studies focused on large-scale models, this work specifically evaluates these mechanisms under constraints representative of real-world edge deployments [5][8]. To determine optimal performance, the methodology employs a comparative experimental design across three conditions: a stateless

baseline, a sparse-retrieval module (BM25 keyword search), and a dense-retrieval module (vector-based semantic search).

1.3 Research Questions

The central problem addressed in this study is:

1. How can a structured text memory be used as a long-term memory mechanism to affect the success of a small language model?
2. How can a vector embedded memory system be used as a long-term memory mechanism to affect the success of a small language model?
3. What are the effects of using both memory architectures on task success rate, memory recall accuracy, memory error rate, retrieval latency, and token utilization?
4. How do the memory architectures differ from each other?

1.4 Research Objectives

By systematically comparing alternative memory designs within a fixed SLM configuration, the study seeks to identify practical and reliable approaches to augment lightweight language models with persistent memory while minimizing latency, hallucinations, and resource overhead. Specifically, the study pursues the following objectives:

1. To design a structured text-based memory mechanism and evaluate its effectiveness as a long-term memory mechanism for SLMs.
2. To design a vector-embedded retrieval memory system and evaluate its effectiveness as a long-term memory mechanism for SLMs.

3. To evaluate the effects of structured text memory and vector-based memory on task success rate, memory recall accuracy, memory error rate, retrieval latency, and token utilization.
4. To compare the two memory architectures across all evaluation dimensions: task success rate, memory recall accuracy, memory error rate, retrieval latency, and token utilization to determine the practical strengths and limitations of each design for SLM-based edge deployment.

1.5 Significance of the Study

LTM is essential for models to maintain continuity across interactions and perform tasks spanning multiple sessions. Without persistent memory, a personal assistant deployed on a local edge device faces a fundamental continuity problem: each new session begins with no knowledge of prior interactions. A user who previously shared their preferences, ongoing tasks, or relevant background information must repeat that context entirely or receive responses that ignore it. This degrades user experience, increases the cognitive burden on the user, and undermines the practical utility of the assistant across multi-session use. For SLMs operating under edge constraints, this problem is compounded: the model cannot rely on large context windows to buffer session history, nor on cloud infrastructure to maintain persistent state. These constraints make the design of an efficient external long-term memory mechanism not merely beneficial but necessary for any SLM-based personal assistant to remain useful across time.

Numerous LTM architectures have been proposed for LLMs, but these designs typically assume that the underlying model possesses sufficient reasoning capacity to filter noisy retrieved information, large context windows to integrate multiple memory entries, and ample computational resources to manage complex retrieval pipelines. These assumptions are often invalid for SLMs deployed in constrained environments, where performance degrades when the retrieval context becomes

noisy or excessively long.

This study addresses these limitations by providing the first systematic empirical evaluation of long-term memory mechanisms designed specifically for SLMs under edge deployment constraints. The findings are directly relevant to researchers developing lightweight AI systems, to practitioners deploying conversational agents in resource-constrained environments such as local edge devices, and to policymakers in developing economies where cloud-based LLM deployment is cost-prohibitive. By identifying which memory architecture achieves the optimal balance of recall accuracy, task success, and latency within strict resource bounds, this study provides actionable deployment guidance for the next generation of private, on-device AI assistants.

1.6 Scope and Limitations

The scope of this study is confined to the integration and evaluation of LTM mechanisms within SLMs deployed in resource-constrained environments. The research focuses exclusively on two external memory designs: structured text-based memory (BM25 Lexical Retrieval) and vector-embedded retrieval memory (Dense Semantic Retrieval) and examines their effects on session-based task performance, contextual continuity, and system efficiency. All experiments are conducted with a fixed SLM configuration to ensure that performance variations are primarily attributable to differences in memory architecture rather than changes in model parameters. The evaluation is limited to textual interaction scenarios that simulate multi-session agent tasks requiring information retention and recall. This study does not include LLMs, multimodal memory systems, or large-scale distributed memory infrastructures, as these approaches require computational resources beyond the intended deployment settings of lightweight agents. It also does not include a direct comparative evaluation between LLM-based and SLM-based memory systems. This delimitation is intentional: the study’s contribution lies in characterizing how LTM behaves within the specific resource constraints of

SLMs, rather than producing a general LLM-vs-SLM benchmark. The research gap is established by prior literature demonstrating that existing LTM frameworks were designed for and evaluated on LLMs [1][2][3], not by empirical comparison within this study.

It also excludes domain-specific fine-tuning, online continual learning, and real-time human-in-the-loop adaptation, as the focus is on external LTM augmentation rather than model retraining. Additionally, the analysis is restricted to controlled experimental environments, which may not fully capture the variability of real-world deployments.

A further limitation of this study pertains to the evaluation scope and intended use case of the augmented model. Due to strict computational and temporal constraints—specifically, local execution on a resource-constrained edge device—it is infeasible to evaluate the model across the entire 23-dataset, 14-domain suite provided by the SLM-Bench framework. Consequently, this study restricts its evaluation to the framework’s “Open-domain” category, focusing exclusively on general question answering and reasoning tasks, alongside personalized, multi-turn conversational datasets. While these selected datasets successfully test the model’s ability to maintain context and reason across disconnected sessions, this targeted scope means the study cannot claim broad domain generalization for the memory architecture in highly specialized fields such as mathematics, legal analysis, or chemistry.

These limitations are acknowledged to maintain a focused investigation on optimizing LTM mechanisms specifically for SLMs operating under constrained computational and deployment conditions.

Chapter 2:

REVIEW OF RELATED LITERATURE AND STUDIES

The rapid evolution of Large Language Models (LLMs) has shifted the research focus from simply scaling parameters to optimizing the information provided during inference—a discipline recently formalized as “Context Engineering” [12]. While foundational models have demonstrated remarkable reasoning capabilities, they remain fundamentally constrained by fixed context windows and static parametric knowledge. To address these limitations, Retrieval-Augmented Generation (RAG) has emerged as the standard paradigm, evolving from “Naive” retrieve-read workflows to “Modular” systems capable of iterative reasoning and tool use [15].

However, a critical dichotomy exists in the current literature. The majority of memory augmentation frameworks such as MemGPT [5] and HippoRAG [7] are designed for and evaluated on massive frontier models (e.g., GPT-4), relying on vast computational resources and high-latency reasoning chains. Conversely, the Small Language Model (SLM) revolution exemplified by Microsoft’s Phi-3 [9] and Google’s Gemma 3 [10] demonstrates that models under 5 billion parameters can rival larger counterparts when data quality is optimized. Yet, research at the intersection of long-term memory (LTM) specifically designed and evaluated for SLMs remains sparse.

Within the Retrieval-Augmented Generation (RAG) paradigm, long-term memory (LTM) refers to the persistent, cross-session storage of information that extends beyond what fits within a model’s finite context window at any given inference step. Du et al. [16] establish a unified taxonomy that divides AI memory into three representational forms: parametric memory, comprising stable knowl-

edge implicitly embedded within model weights and accessed through feedforward computation at inference; contextual unstructured memory, comprising explicit external data in raw text formats such as conversation histories and retrieved passages; and contextual structured memory, comprising explicit external data organized in symbolic formats such as knowledge graphs or relational records. Temporally, Du et al. [16] further distinguish between long-term memory as persistent storage spanning multiple sessions, including multi-turn dialogue histories and accumulated user-specific facts, and short-term memory, corresponding to the transient, session-level context currently present in the active context window. In an applied RAG framework, LTM is operationalized as external contextual memory: prior interactions and relevant facts are written to an external store and selectively retrieved at query time through retrieval and compression operations, then injected into the model’s context window to ground generation in accumulated session history. For SLMs, this external memory pipeline carries particular risk alongside its benefits: the model’s limited reasoning bandwidth means that noisily retrieved or poorly compressed memories can degrade rather than improve response quality—a failure mode that motivates the controlled retrieval architecture comparison at the center of this study.

This chapter synthesizes the literature across four thematic pillars to contextualize this gap. First, it examines memory architectures for large language models, covering both internal context window optimization and external memory augmentation strategies. Second, it analyzes vector-embedded memory systems, tracing the evolution of RAG from static pipelines to dynamic, production-ready deployments. Third, it reviews memory organization strategies including topic-based, type-based, dynamic network, and cognitive forgetting-curve approaches and the broader discipline of context engineering. Fourth, it evaluates the capabilities and constraints of small language models, focusing on the architectural innovations that make on-device memory systems theoretically feasible. By contrasting the heavy, OS-inspired memory systems of the LLM era with the efficiency

requirements of the Edge AI era, this review establishes the necessity for a specialized memory framework enabling SLMs to maintain continuity across disconnected sessions without exceeding the resource constraints of local deployment.

2.1 Memory Architectures for Large Language Models

Memory architectures determine how language models store, organize, and retrieve contextual information across interactions. Two principal strategies have emerged: optimizing the model’s internal context window, and augmenting the model with external memory storage. Each approach carries distinct trade-offs in terms of capacity, computational overhead, and suitability for cross-session persistence.

2.1.1 Internal Context Window Optimization

A central objective of recent LLM development has been extending the practical boundaries of context windows so that models can process longer inputs without losing coherent recall of earlier content. The Gemma 3 Technical Report [10] pursues this objective through architectural redesign. The Gemma 3 family introduces lightweight models ranging from 1B to 27B parameters with contexts up to 128K tokens. Crucially, the 4B variant employs a 5:1 interleaving of local sliding window and global attention layers, reducing KV-cache memory by approximately 85% compared to full global attention.

Evaluation instruments include standard academic benchmarks such as MMLU, HumanEval, and MathVista, on which the 4B model achieves performance competitive with models 6–7 $\times$ larger. The report also employs multi-turn benchmarks such as MRCR.

However, a key limitation of this study is that its evaluation remains confined to static performance metrics within single sessions, rather than measuring cross-session persistence metrics of the kind defined by Hong et al. [15]. This gap is directly relevant to the present study, which seeks to evaluate how Gemma 3 4B’s

hybrid attention interacts with externally injected memory across disconnected sessions.

A complementary approach to expanding context was proposed by Chen et al. [11], whose work on Positional Interpolation aimed to extend context windows of RoPE-based models without architectural overhaul. Their method down-scales position indices to match the original training window rather than extrapolating beyond it. Evaluation on LLaMA models extended from 2K to 32K tokens using only 1,000 fine-tuning steps showed that models maintained performance on original short-context tasks with less than 2% degradation. Theoretical analysis confirmed that the interpolation bound is approximately $600\times$ smaller than extrapolation, explaining the superior stability. Notwithstanding this progress, both Gemma 3’s hybrid attention and the positional interpolation method of Chen et al. [11] are ultimately bounded by finite context windows. As Hong et al. [15] demonstrate, effective utilization falls consistently below theoretical capacity due to the phenomenon they term “Context Rot”—a finding that challenges the premise that expanding context length alone is sufficient for long-term continuity.

2.1.2 External Memory System

Recognizing the ceiling of internal context expansion, several research groups have pursued architectures that supplement a model’s fixed context with external memory retrieval. Wang et al. [4] proposed LONGMEM with the objective of enabling LLMs to access effectively unlimited-length context without catastrophic forgetting or memory staleness. Their system augments a frozen LLM backbone with a trainable SideNet (approximately 37% of backbone parameters) that performs chunk-level memory retrieval.

Evaluation was conducted on the PG-22 dataset of long-form literary text, yielding perplexity reductions of 1.38–1.62 and a 40.5% accuracy on the ChapterBreak benchmark, significantly outperforming the approximately 28% baseline. A notable advantage is the system’s retrofitting capability, requiring no full re-

training of the base model. However, the authors acknowledge significant limitations: the retrieval component introduces computational overhead equivalent to approximately 55% of backbone inference time, and the framework was evaluated exclusively on continuous long-form text sequences (e.g., books) rather than on the disconnected, multi-session dialogue scenarios that characterize real-world personal assistants.

Packer et al. [5] took a different approach, designing MemGPT with the explicit objective of enabling cross-session memory persistence. Inspired by operating system design, MemGPT implements a hierarchical virtual context management system in which the model self-directs memory operations via function calls, moving data between fast in-context storage and slower external storage tiers. The study’s key contribution is quantitative benchmarking of memory consistency and engagement across multi-session agents. Despite demonstrating successful cross-session persistence, a critical limitation emerged: the system relies heavily on the underlying model’s function-calling capabilities. When evaluated with GPT-3.5 rather than GPT-4, the authors observed marked brittleness and performance degradation, suggesting that MemGPT’s architecture may overwhelm the reasoning bandwidth of smaller models. This finding directly motivates the present study. While Wang et al. [4] validate memory augmentation on a small model (407M parameters), their evaluation is limited to continuous sequences; conversely, Packer et al. [5] achieve cross-session persistence but only with large, capable models. A critical gap therefore remains: achieving effective cross-session persistence specifically within the constraints of SLMs, which is precisely the problem this research addresses.

2.2 Vector-Embedded Memory Systems

Vector embeddings form the retrieval backbone of most external memory systems. This section examines how RAG architectures have evolved to meet the demands of knowledge-intensive tasks, how production deployment challenges

have shaped system design, and how specialized applications have demonstrated RAG’s viability even on smaller architectures.

2.2.1 RAG Architecture Evolution

Gao et al. [14] conducted a comprehensive survey whose objective was to systematically map the development of RAG paradigms and identify open challenges. Their taxonomy organizes RAG into three generational stages: (1) Naive RAG, which gained prominence post-ChatGPT and follows a straightforward “Retrieve-Read” framework, often limited by low retrieval precision and recall; (2) Advanced RAG, which introduces pre- and post-retrieval optimization strategies such as query rewriting, re-ranking, and context compression to address indexing and generation difficulties; and (3) Modular RAG, which offers greater adaptability through specialized components including Search, Memory, and Routing modules that allow for flexible orchestration and iterative retrieval flows. The survey identifies a significant open question: while scaling laws are well-established for LLMs, their applicability to RAG systems remains uncertain. Of particular interest to the present work is the survey’s attention to “Inverse Scaling Laws,” which suggests that smaller models may effectively compete with or outperform larger ones in specific retrieval-augmented setups—a hypothesis this thesis investigates empirically through Gemma 3 4B.

The choice of retrieval mechanism within a RAG pipeline has measurable consequences for downstream generation quality. Sawarkar et al. [17] demonstrate this directly through their Blended RAG framework, which evaluates three distinct retrieval strategies — a BM25 index for keyword-based full-text search, a dense vector index for semantic similarity matching, and a sparse encoder index for nuanced term relationship capture blended through hybrid query strategies. Evaluating across the NQ, TREC-COVID, HotPotQA, and SQuAD benchmarks, they find that semantic search-based hybrid queries consistently realize better retrieval precision than conventional keyword or vector-based approaches. Specif-

ically, their blended combination utilizing a Sparse Encoder with a “Best Fields” query strategy consistently set new accuracy benchmarks, surpassing even fine-tuned, domain-adapted retrieval performance on SQuAD in a zero-shot setting. Critically, their results show that retriever selection is the single most consequential design decision in a RAG system, as efforts to optimize the generator (LLM) yield limited impact if the retriever feeds irrelevant context. This finding motivates the controlled comparison at the center of the present study: rather than assuming dense retrieval universally dominates, this study empirically isolates BM25 lexical retrieval and dense semantic retrieval as independent experimental conditions to characterize their relative performance under SLM and edge deployment constraints.

2.2.2 Production Deployment and Latency Optimization

Beyond architectural design, the practical viability of RAG in production environments depends critically on latency. Shen et al. [18] systematically decomposed RAG inference latency with the objective of identifying the dominant sources of overhead and guiding production optimization. Their analysis employed a suite of indexing strategies and batch sizes as experimental instruments, finding that with aggressive re-retrieval (stride length of 4), RAG components account for 97% of total inference latency with retrieval contributing 36% and prefill contributing 45%. Among indexing strategies, HNSW-SQ prioritizes recall with latencies starting at approximately 310ms, whereas IVF-SQ permits lower latencies starting at approximately 170ms but scales more poorly with batch size. A particularly salient finding is that while a standard LLM achieves a Time-To-First-Token of approximately 495ms, adding RAG more than doubles this to approximately 965ms. This overhead directly informs the latency budget defined in the present study, specifically, the target of sub-500ms retrieval latency identified as necessary for non-disruptive conversational interactions.

Complementing this system-level analysis, Gan et al. [19] surveyed 582 pa-

pers with the objective of establishing a unified evaluation taxonomy for RAG systems, covering retrieval quality, generation quality, safety, and efficiency. The survey confirms that traditional datasets such as Natural Questions, HotpotQA, and MS MARCO remain dominant evaluation standards. Critically, it identifies a gap directly relevant to this research: the lack of standardized benchmarks for memory persistence across disconnected sessions. While emerging benchmarks like MTRAG (2025) begin to address multi-turn capabilities, the majority of evaluations still focus on continuous contexts or single queries, failing to capture retrieval accuracy degradation over long-term, interrupted interactions. The present study’s dynamic disconnected-session protocol is partly motivated by this absence.

Addressing the practical question of whether memory systems can meet production-grade performance requirements, Chhikara et al. [20] introduced Mem0, a scalable memory-centric architecture that dynamically extracts, consolidates, and retrieves salient information from ongoing multi-session conversations. The system was evaluated comprehensively on the LOCOMO benchmark [21] against six baseline categories, including established memory systems, RAG variants, full-context approaches, and proprietary solutions. Mem0 achieves 26% relative improvement over the OpenAI proprietary memory baseline on the LLM-as-a-Judge metric, and its graph-enhanced variant (Mem0g) achieves a further 2% overall score improvement. Crucially for edge-deployment considerations, Mem0 attains 91% lower p95 latency and reduces token costs by over 90% compared to the full-context baseline, demonstrating that structured memory retrieval offers a compelling balance between accuracy and computational load. This finding reinforces the thesis’s central argument that external memory systems are superior to context window expansion for resource-constrained deployment, and establishes concrete efficiency benchmarks that the present study can reference in evaluating the Gemma 3 4B configuration.

2.2.3 Specialized RAG Applications

Gutiérrez et al. [7] introduced HippoRAG with the objective of enabling single-step multi-hop reasoning in RAG systems, inspired by the hippocampal indexing theory of human memory. Their system uses a schemaless knowledge graph and Personalized PageRank (PPR) for retrieval. Evaluated on the 2Wiki-MultiHopQA benchmark, HippoRAG achieves 89.5% Recall@5, significantly outperforming the ColBERTv2 baseline (68.2%), while maintaining competitive performance on MuSiQue and HotpotQA. In terms of computational efficiency, HippoRAG is 10–30× cheaper and 6–13× faster than iterative multi-hop methods such as IRCoT. Importantly for this study, the framework was evaluated on both closed frontier models (GPT-3.5) and SLMs (Llama-3.1-8B), confirming its viability on smaller architectures. Furthermore, unlike static summarization approaches, its graph-based design explicitly supports continuous knowledge integration by adding new nodes and edges without requiring full re-indexing—a property particularly relevant for multi-session memory systems where new interactions must be incorporated incrementally.

2.3 Memory Organization Strategies

Beyond storage capacity and retrieval speed, the organizational structure of a memory system determines how effectively relevant information can be surfaced in context. This section examines four organizational approaches: topic-based summarization, type-based multi-agent architectures, dynamic network evolution, and cognitively-inspired forgetting mechanisms, as well as the overarching framework of context engineering that situates these approaches within a unified discipline.

2.3.1 Topic-Based and Type-Based Organization

Tan et al. [2] proposed Reflective Memory Management (RMM) with the objective of improving long-term personalized dialogue through principled memory

organization. The system employs a dual strategy: prospective topic-based summarization organizes incoming information as it is received, while retrospective RL-refined retrieval uses online reinforcement learning with unsupervised LLM attribution signals as self-supervised rewards to optimize memory access patterns over time. Evaluation was conducted on the LongMemEval benchmark [6], where RMM achieves up to 73.8% accuracy—significantly outperforming standard RAG baselines at approximately 64%—and on the MSC dataset for multi-session consistency. A notable advantage is the elimination of costly labeled training data through self-supervised attribution. However, the authors acknowledge limitations: the RL training introduces computational overhead, and the quality of generator feedback affects retrieval refinement. Importantly, RMM explicitly addresses multi-session continuity via the MSC benchmark, making it one of the few frameworks to evaluate the scenario directly relevant to this thesis. Its recommendation to combine prospective topic organization with retrospective retrieval refinement informs the memory management design considered in the present study.

Wang and Chen [1] pursued a different organizational philosophy with MIRIX, introducing a modular multi-agent memory architecture whose objective was to mirror the diversity of human memory types. The system orchestrates six specialized memory components: Core, Episodic, Semantic, Procedural, Resource, and Knowledge Vault, each managed by a dedicated agent. Evaluated on the LOCOMO benchmark [21], MIRIX achieves a state-of-the-art 85.4% accuracy, far surpassing the standard RAG baseline (51.6%) and competing memory systems such as Zep (79.1%). The architecture’s explicit typing of memory categories enables specialized retrieval strategies for different information types, which is a clear improvement over undifferentiated vector stores. However, the system introduces significant architectural complexity, requiring eight distinct agents and models with strong function-calling capabilities to manage routing and updates effectively. This dependency on complex function-calling is a critical limitation for SLMs, as the present study notes that models under 5B parameters may lack

the reasoning bandwidth to reliably orchestrate such multi-agent pipelines.

A contrasting perspective on complexity is offered by Patel and Patel [22], who proposed ENGRAM (Effective, Lightweight Memory Orchestration for Conversational Agents) with the explicit objective of demonstrating that architectural simplicity can achieve competitive or superior performance to complex multi-stage memory systems. ENGRAM organizes conversation into three canonical memory types: episodic, semantic, and procedural, through a single router and retriever, converting each user turn into typed memory records with normalized schemas and dense embeddings. At query time, the system retrieves top-k dense neighbors per type and merges results with simple set operations. The central claim of the study is that “careful memory typing, minimal routing, and straightforward retrieval suffice” to achieve state-of-the-art performance. Evaluated on LoCoMo and LongMemEval [6], ENGRAM achieves state-of-the-art performance on LoCoMo, excelling in single-hop, multi-hop, and open-domain categories, and surpasses full-context baselines on LongMemEval by 15 absolute points (71.4% vs 56.2%). These results were achieved without knowledge graphs, multi-stage retrieval pipelines, or OS-style schedulers. This finding directly supports the present study’s architectural philosophy: rather than implementing the autonomous OS-style memory management of MemGPT [5] or the eight-agent orchestration of MIRIX [1], this paper investigates whether a simpler, decoupled retrieval policy can achieve sufficient cross-session performance within the resource constraints of Gemma 3 4B.

2.3.2 Dynamic Knowledge Networks and Cognitive Memory Models

Xu et al. [3] proposed A-MEM (Agentic Memory) with the objective of moving beyond static knowledge storage toward a system in which memory actively evolves through use. Drawing on Zettelkasten principles, A-MEM creates atomic notes with autonomous linking and dynamic evolution. Unlike systems relying on predefined schemas or rigid graph databases, A-MEM employs a dual-process

mechanism: Link Generation, in which the LLM identifies shared attributes to connect new memories with existing ones, and Memory Evolution, in which new experiences trigger retroactive updates to existing contextual descriptions. Of particular significance to this research, Xu et al. [3] explicitly evaluated A-MEM on SLMs. On the LoCoMo benchmark, Llama 3.2 1B and Qwen 2.5 1.5B equipped with A-MEM significantly outperformed baselines such as MemGPT and MemoryBank. For instance, Llama 3.2 1B with A-MEM achieved a ROUGE-L score of 60.23 on adversarial tasks compared to MemGPT’s 50.31. To address scalability concerns, the system was stress-tested with memory repositories ranging from 1,000 to 1,000,000 entries, maintaining exceptional retrieval efficiency with latency increasing only from $0.31\ \mu\text{s}$ to $3.70\ \mu\text{s}$. The authors acknowledge a key limitation: “the quality of memory evolution is influenced by the inherent capabilities of the underlying language models,” meaning smaller models may still generate lower-quality contextual descriptions. This unresolved limitation represents a gap this thesis aims to characterize empirically for the Gemma 3 4B architecture.

A complementary line of research has approached memory management from a cognitive science perspective. Zhong et al. [12] proposed MemoryBank with the objective of creating a long-term memory mechanism grounded in established psychological theory of human forgetting. The system builds on the Ebbinghaus Forgetting Curve, which describes how memory retention decays exponentially over time unless reinforced. MemoryBank translates this principle into a dynamic memory updating mechanism: memories are selectively forgotten or strengthened based on elapsed time and the frequency with which they are recalled during subsequent interactions. The architecture combines a memory storage component, a dense-retrieval-based memory retriever, and a personality-aware memory updater that builds evolving user profiles from interaction history. The system was applied to create SiliconFriend, an LLM-based companion chatbot, and evaluated through both qualitative analysis on real-world user dialogues and quantitative analysis using simulated multi-topic interactions. Results demonstrate that the

forgetting-curve mechanism produces more natural, personalized long-term interactions compared to static storage. MemoryBank’s contribution to the present study is conceptual: its demonstration that selective forgetting is not a failure mode but a desirable design property—particularly for edge deployments where storage capacity is limited—informs how the present research frames the problem of memory management across disconnected sessions for Gemma 3 4B.

2.3.3 Context Engineering Framework

Mei et al. [8] formalized the discipline of Context Engineering with the objective of providing a unified theoretical and practical framework for the field. Surveying over 1,400 papers, they establish a taxonomy covering three foundational components: Retrieval/Generation, Processing, and Management. Their analysis reveals that while Context Processing using attention mechanism innovations like Ring Attention or LongNet addresses the $O(n^2)$ computational complexity of Transformers, it does not solve the “lost-in-the-middle” phenomenon, where performance degrades by up to 73% in long contexts. Consequently, Mei et al. [8] emphasize Context Management, and specifically Memory Hierarchies, as the critical solution for persistent interaction. They categorize memory systems into OS-inspired architectures (like MemGPT [5]) that page information between context and external storage, and Dynamic organizations (like MemoryBank [12]) based on cognitive forgetting curves. Their key finding—that a “fundamental asymmetry” exists in which models are better at understanding context than generating long-form outputs based on it—provides an important constraint for memory system design: retrieved context must be well-structured to yield high-quality generation, particularly in resource-constrained models where this asymmetry may be more pronounced.

2.3.4 BM25 Lexical Retrieval vs Dense Semantic Retrieval

A fundamental architectural distinction exists between BM25 lexical retrieval and dense semantic retrieval, each representing divergent approaches to information storage and access within a RAG memory backend. BM25 lexical retrieval maintains memory records as human-readable text, applying a probabilistic term-frequency scoring function to match queries against stored records through an inverted index — enabling direct inspection, rule-based access, and exact-term matching without any embedding computation. In contrast, dense semantic retrieval encodes memory records as high-dimensional numerical vectors, enabling similarity matching through geometric distance metrics at the cost of interpretability and continuous embedding computation overhead.

The sparse retrieval backend evaluated in this study is grounded in the BM25 probabilistic ranking function, whose theoretical foundations were formally established by Robertson and Zaragoza [23]. BM25 operates by scoring documents against a query using three weighted components: term frequency within a document, inverse document frequency across the corpus, and a document length normalization factor that prevents longer documents from receiving inflated scores solely due to size. Unlike earlier TF-IDF approaches, BM25 incorporates a term frequency saturation parameter that prevents any single high-frequency term from dominating the relevance score, and a document-length normalization parameter that can be tuned to the characteristics of the target corpus. Robertson and Zaragoza [23] situate BM25 within the broader probabilistic relevance framework, establishing it not as a heuristic but as a principled estimation of the probability that a document is relevant to a given query. This theoretical grounding has made BM25 the default sparse retrieval standard in information retrieval for over a decade, serving as the baseline against which neural and dense retrieval methods are routinely benchmarked. For the present study, BM25 is implemented through SQLite FTS5’s inverted index architecture, which applies the same probabilistic

term-frequency scoring to memory records stored across sessions. For the present study, BM25 is implemented through SQLite FTS5’s inverted index architecture, which applies the same probabilistic term-frequency scoring to memory records stored across sessions. The practical implications of this retrieval approach become clearer when examined through existing implementations of structured text memory in the literature — most notably MemGPT, which reveals both the interpretability advantages and the reasoning-bandwidth costs that lexical memory management imposes on the underlying model. u

Packer et al. [5] exemplify structured text memory through MemGPT’s “edit buffer” mechanism, which maintains a working context of structured text including system instructions, user messages, and conversation summaries. The system manages this structured buffer through explicit function calls (`send_message`, `core_memory_append`, `core_memory_replace`), allowing the model to self-direct memory operations via symbolic manipulation rather than semantic search. This approach provides interpretability to developers and users who can directly inspect exactly what information resides in memory and how it is organized, but requires the model to possess sufficient reasoning capacity to explicitly manage insertion, updating, and deletion operations through deliberate cognitive steps. The brittleness observed when evaluating MemGPT with GPT-3.5 versus GPT-4 suggests that structured text memory management may exceed the reliable reasoning bandwidth of smaller models, a limitation central to the present study’s investigation.

Tan et al. [2] demonstrate another form of structured text memory through RMM’s topic-based summarization, where conversations are distilled into text summaries organized by topic categories. Unlike embedding-based retrieval that depends on continuous semantic similarity scores computed through vector operations, RMM’s retrieval mechanism selects summaries based on explicit topic alignment determined through retrospective analysis and reinforcement learning refinement. This structured approach achieved 73.8% accuracy on LongMemEval

compared to approximately 64% baseline, demonstrating that principled text organization can outperform unstructured storage. However, the acknowledged computational overhead for summary generation and RL training highlights the trade-off between retrieval precision and resource cost—particularly relevant for edge deployment scenarios.

Zhong et al. [12] propose MemoryBank, which maintains structured text-based user personality profiles alongside conversation history. The system stores explicit attributes (preferences, facts, relationships) as text records rather than embeddings, enabling direct inspection and modification without requiring vector similarity computation. Memory decay is implemented through time-based weights applied to text records rather than similarity-based filtering, aligning with the Ebbinghaus Forgetting Curve principle where retention decreases exponentially unless reinforced through recall. This approach provides cognitive alignment—the memory management mechanism mirrors human memory processes—but evaluation remains qualitative through the SiliconFriend chatbot without quantitative benchmarking on standardized datasets.

In contrast, pure vector-embedded approaches prioritize semantic retrieval over structural organization. Gutiérrez et al. [7] demonstrate this through HippoRAG, which synergizes dense retrieval with knowledge graphs. Unlike pure vector baselines, HippoRAG constructs a schemaless graph from passages and utilizes the Personalized PageRank algorithm to simulate associative memory traversal. This hybrid approach achieves 89.5% Recall@5 on 2WikiMultiHopQA, significantly outperforming standard RAG methods. Wang et al. employ LONGMEM’s SideNet to encode memory chunks as cached key-value vectors, retrieving relevant context through cosine similarity scores without requiring the model to manage memory organization explicitly. Chhikara et al. extend these principles through Mem0, which achieves a 91% reduction in p95 latency through optimized vector indexing and graph-based consolidation, demonstrating that retrieval-centric architectures can simultaneously improve both accuracy and efficiency.

The trade-offs between these approaches remain empirically understudied, particularly for small language models operating under resource constraints. Structured text memory offers several theoretical advantages: interpretability (humans can inspect and verify stored content), explicit control (deterministic retrieval through rule-based filtering), and potentially lower computational overhead during retrieval since no embedding generation is required at query time. However, these advantages depend critically on the model’s reasoning capacity to manage memory operations effectively—inserting relevant information, updating outdated records, and retrieving appropriate context through symbolic reasoning rather than semantic matching. If the model lacks sufficient reasoning capacity, structured memory management may introduce errors, inconsistencies, or retrieval failures.

Vector-embedded retrieval provides complementary strengths: robust semantic matching without requiring explicit memory management reasoning (the model need not understand how to organize or filter memory, only how to generate queries), automatic generalization across paraphrases and synonyms through semantic space proximity, and proven scalability to millions of entries as demonstrated by A-MEM [3] with retrieval latency increasing only from $0.31\ \mu\text{s}$ to $3.70\ \mu\text{s}$ across 1,000 to 1,000,000 entries. However, these advantages introduce distinct costs: embedding computation overhead (each memory insertion and query requires neural encoding), lack of interpretability (dense vectors are opaque to human inspection), and dependency on embedding model quality (poor embeddings yield poor retrieval regardless of memory organization).

For hybrid-attention SLMs like Gemma 3 4B operating under edge deployment constraints, the question of whether structured text or vector-embedded memory better supports cross-session continuity remains empirically open. While Terranova et al. [24] confirm that smaller foundation models benefit most from simpler architectures, they do not distinguish between text-based simplicity (minimal symbolic reasoning requirements) and embedding-based simplicity (minimal architectural complexity). The capability-dependency limitation identified by Xu

et al. [3]—that memory quality depends on the underlying model’s reasoning capacity—suggests that structured text memory may be vulnerable to errors when deployed on SLMs with limited reasoning bandwidth, yet this hypothesis has not been empirically tested on hybrid-attention models under resource constraints.

The selection of the embedding model for the dense retrieval backend requires explicit justification, as the choice directly shapes retrieval quality, memory overhead, and inference feasibility under edge constraints. This study employs EmbeddingGemma (300M parameters), a lightweight open text embedding model built on the Gemma 3 language model family, trained using a recipe that combines encoder-decoder initialization, geometric embedding distillation, and checkpoint merging across varied data mixtures [25]. Its selection over widely-used alternatives such as all-MiniLM-L6-v2 (22.7M parameters, 384 dimensions) is motivated by three considerations. First, architectural coherence: EmbeddingGemma shares the same foundational model family and training lineage as Gemma 3 4B, meaning the semantic space in which memories are encoded is closely aligned with the semantic space in which the generative model processes queries, reducing the representational mismatch that can degrade retrieval relevance when embedding and generation models are trained on divergent corpora. Second, MRL-enabled dimensionality reduction: Vera et al. [25] explicitly demonstrate that EmbeddingGemma retains competitive performance when truncating embedding outputs, enabling reduction from 768 to 128 dimensions and reducing FAISS index storage requirements by approximately 83%—a reduction critical for sustaining sub-4GB total memory usage across simultaneous embedding, retrieval, and generation on edge hardware. Third, verified on-device suitability: Vera et al. [25] designed EmbeddingGemma explicitly for low-latency and high-throughput use cases including on-device applications, and demonstrate state-of-the-art results on the Massive Text Embedding Benchmark (MTEB) across English, multilingual, and code domains, outperforming both proprietary and open models under 500M parameters—confirming that the model’s retrieval quality does not come

at the cost of the computational efficiency this study requires.

Furthermore, the latency implications differ fundamentally between approaches. Structured text memory avoids embedding computation during retrieval, potentially enabling sub-500ms latency targets through simple rule-based filtering (e.g., retrieve last N turns or messages within time window T). However, this assumes the model can generate appropriate symbolic queries, which may require additional reasoning tokens. Vector-embedded memory incurs embedding computation overhead but benefits from highly optimized HNSW or IVF indexing as demonstrated by Shen et al. [18], achieving retrieval times of 170–310ms on server-grade hardware. Whether these latency characteristics translate to edge deployments with sub-4GB RAM and battery-limited cycles remains uncharacterized.

The present study addresses this gap by empirically comparing structured text memory (rule-based retrieval with explicit symbolic organization) and vector-embedded retrieval (semantic similarity matching with dense indexing) for Gemma 3 4B across disconnected sessions under explicit edge constraints. This comparison provides the first controlled empirical evidence of whether SLMs benefit more from interpretable structured text or robust semantic matching, informing practical deployment decisions for resource-constrained personal assistants.

2.4 Evaluation Methodologies for Memory Systems

Designing memory systems requires equally rigorous evaluation frameworks. This section examines the benchmarks and methodologies that have been used to assess memory-augmented language models, with particular attention to the LongMemEval benchmark, the phenomenon of context rot, and the distinction between theoretical context capacity and effective context utilization.

2.4.1 Context Understanding and Memory Benchmarks

The selection of evaluation benchmarks reflects different research priorities in the field. Wu et al. [6] introduced LongMemEval with the objective of providing a

comprehensive and scalable benchmark specifically designed to evaluate the long-term memory capabilities of chat assistants across five core abilities: information extraction, multi-session reasoning, temporal reasoning, knowledge updates, and abstention. With 500 meticulously curated questions embedded within freely scalable user-assistant chat histories, LongMemEval presents a significant challenge to existing systems: commercial chat assistants and long-context LLMs show a 30–60% accuracy drop on the benchmark under extended interaction conditions.

The framework decomposes memory design into three stages: indexing, retrieval, and reading. While Wu et al. demonstrate that optimizations like fact-augmented key expansion and time-aware query expansion improve recall, their evaluation remains limited to static, pre-recorded conversation templates. This represents a critical methodological gap: the benchmark fails to account for the dynamic context drift and temporal gaps inherent in production edge deployments. Consequently, this study adopts the LongMemEval taxonomy but applies it to a dynamic disconnected-session protocol. By doing so, it tests whether Gemma 3 4B’s hybrid-attention mechanism can maintain the “Reading” stability (Stage 3) required to overcome the “Context Rot” documented in larger frontier models. LongMemEval is accepted at ICLR 2025 and its benchmark and code are publicly available, making it the evaluation instrument of choice for RMM [12] and a relevant standard for this thesis.

Beyond LongMemEval, other benchmarks address complementary evaluation dimensions. Tan et al. [2] employ the MSC dataset to evaluate conversational quality via METEOR and BERT scores across disconnected chats. Wang and Chen [1] validate MIRIX on the LOCOMO benchmark [21], which focuses on long-term conversational memory across extended interaction histories, while Wang et al. [4] employ ChapterBreak, where their LONGMEM model achieves 40.5% accuracy against a sub-30% baseline. For multi-hop reasoning, Gutiérrez et al. [7] report that HippoRAG with IRCoT achieves 82.3% Recall@5 on 2WikiMultiHopQA. Analysis of these benchmarks reveals a consistent limitation identified by Tan et

al. [2]: most systems rely on fixed retrieval granularity—for example, retrieving whole chunks or sessions—rather than adapting to the diverse, semantically disconnected structures of real-world long-term interactions. The present study addresses this limitation by establishing a dynamic disconnected-session protocol that simulates realistic temporal gaps and context drift.

2.4.2 Context Rot Phenomenon

Hong et al. [15] conducted one of the most rigorous analyses of long-context model behavior, examining 18 models including GPT-4.1, Claude 4, Gemini 2.5, and Qwen3 with the objective of determining how and why model performance degrades as input length increases.

Using an extended Needle-in-a-Haystack (NIAH) protocol enriched with semantic variations as the primary evaluation instrument, their study reveals three principal findings: (1) all evaluated models exhibit non-uniform performance degradation as input length increases; (2) needle-question semantic similarity is a critical factor—performance degrades significantly faster when the needle is semantically ambiguous relative to the query compared to high-similarity exact matches; and (3) the presence of distractors compounds this decay, with even four distractors causing notable hallucinations particularly in GPT models, while Claude models tended to abstain rather than confabulate. Contrary to the standard “Lost in the Middle” hypothesis, Hong et al. [15] find no notable performance variation based on needle position for retrieval tasks; instead, degradation is driven by semantic ambiguity and haystack structure, with coherently structured haystacks actually performing worse than shuffled ones. The authors recommend isolating input length as an experimental variable in benchmarks to distinguish between task complexity effects and context-length effects, and conclude that standard lexical NIAH benchmarks are insufficient to capture real-world degradation patterns. For the present study, this implies that simply extending Gemma 3 4B’s context window is not a viable solution to cross-session continuity—a finding that

provides empirical justification for the external retrieval-based approach adopted here.

2.4.3 Memory Architecture Scaling with Model Capability

A critical empirical question for this thesis is whether memory architectures designed for large models transfer their benefits to smaller ones. Terranova et al. [24] addressed this directly through a systematic evaluation of memory-augmented methods on long-context dialogues requiring diverse reasoning strategies. Their study examined full-context prompting, semantic memory through both RAG and agentic memory (A-MEM [3]), episodic memory through in-context learning, and procedural memory through prompt optimization evaluated consistently across multiple LLMs of varying capability using the LoCoMo benchmark. The key finding is directly relevant to this thesis: memory-augmented approaches reduce token usage by over 90% while maintaining competitive accuracy, and memory architecture complexity should scale with model capability—specifically, smaller foundation models benefit most from RAG, while stronger instruction-tuned models can additionally leverage more complex episodic and agentic semantic memory. The authors further find that episodic memory helps LLMs recognize the limits of their own knowledge, mitigating the metacognitive limitations identified elsewhere in the literature. This result provides empirical grounding for the present study’s hypothesis that a relatively simple retrieval-based memory framework may be the optimal design for Gemma 3 4B, rather than the more complex autonomous architectures validated primarily on larger models.

2.5 Small Language Models: Capabilities and Constraints

The feasibility of deploying memory-augmented language models on edge devices depends on whether SLMs can serve as capable reasoning engines while operating within strict resource budgets. This section examines the architectural and training innovations that have enabled recent SLMs to achieve performance

once associated exclusively with much larger models.

2.5.1 Data Quality Over Model Scale

Microsoft [9] introduced Phi-3-mini (3.8B parameters) to challenge scaling laws, demonstrating that data quality—specifically 3.3 trillion tokens of “textbook-quality” synthetic and filtered data—can enable reasoning capabilities competitive with GPT-3.5. Phi-3-mini achieves 68.8% on MMLU and 82.5% on GSM8K, suggesting that SLMs can internalize complex logic within a compact parameter space. However, this “reasoning density” creates a false sense of reliability for edge deployment. While architectural innovations like Grouped-Query Attention (GQA) in the 7B variant optimize KV-cache efficiency, the model’s performance remains tethered to its internal parametric state.

The critical limitation of the Phi-3 report is its reliance on static benchmarks like RULER to validate long-context capabilities. These evaluations measure the model’s ability to process a single, continuous stream of data rather than its capacity to maintain cross-session persistence. The data quality innovations that enable Phi-3’s single-session reasoning have not been stress-tested against the noise and context drift of external memory systems. This study addresses this gap by investigating whether the reasoning density of a similar-scale model, Gemma 3 4B, is sufficient to manage the “Reading” stage of an external memory framework when subjected to a dynamic disconnected-session protocol.

The Gemma 3 [10] family spans 1B to 27B parameters, demonstrating that specialized competencies can be distilled into SLMs through focused training objectives. However, the primary technical justification for utilizing the Gemma 3 4B model lies in its aggressive architectural optimization of the KV-cache. Unlike the dense attention mechanisms of the Phi-3 or Llama families, Gemma 3 4B employs a 5:1 interleaving of local sliding window attention and global attention layers. This hybrid structure reduces KV-cache memory consumption by approximately 80% compared to full global attention, theoretically enabling the model to handle

the expanded context required for external memory injection without exceeding mobile-class VRAM limits.

Yet, this efficiency introduces a significant reasoning constraint: since only a fraction of the layers maintain a global view, the model’s ability to “attend” to retrieved long-term memories may be structurally compromised compared to larger, dense-attention models. This study investigates this trade-off, specifically testing if the 5:1 hybrid architecture can maintain the reasoning bandwidth necessary to synthesize disconnected session data at the “Reading” stage.

2.6 Synthesis and Research Justification

The preceding review establishes a fundamental tension between the increasing reasoning density of SLMs and the catastrophic performance degradation characterized as “Context Rot” in extended interactions. While architectural innovations such as Gemma 3 4B’s 5:1 hybrid attention layers significantly reduce memory overhead, the literature reveals a persistent Reasoning-Resource Mismatch: complex memory frameworks like MemGPT or MIRIX often exceed the orchestration capabilities of models under 5B parameters.

This synthesis identifies three critical gaps: (1) the lack of optimization for hybrid-attention architectures in LTM frameworks, (2) the reliance on static benchmarks rather than dynamic, disconnected-session protocols, and (3) the absence of a defined Pareto frontier for accuracy and retrieval latency on edge hardware. By pivoting from autonomous OS-style management to efficient, decoupled retrieval, this research justifies a specialized framework designed to maintain persistent personalization within the strict VRAM and reasoning constraints of the Gemma 3 4B architecture.

2.6.1 Convergent Findings Across Literature

Analysis of the reviewed literature reveals five strong points of convergence, as detailed comprehensively in Section 2.7.5. First, external memory systems con-

sistently improve performance over non-augmented baselines across all evaluated frameworks. Second, memory organization matters beyond raw storage capacity, with structured approaches outperforming flat vector retrieval. Third, sub-500ms retrieval latency is achievable through index optimization. Fourth, context utilization consistently falls below advertised context capacity due to the Context Rot phenomenon [15]. Fifth, data quality enables competitive small model performance [9]. These convergent findings establish the empirical foundation for memory-augmented SLMs.

The literature presents productive tensions that resolve upon examination. The tension between context window growth [6] [9] and context rot [15] resolves when recognizing these as complementary: internal optimization suits single-session reasoning while external retrieval enables cross-session persistence. The tension between architectural complexity [6] and simplicity [22] suggests that complexity should be added only when empirically justified—a principle guiding the present study’s design. Detailed analysis of these tensions appears in Section 2.7.5.

The literature presents several productive tensions that necessitate the specific architectural choices of this study:

Context Capacity vs. Effective Retrieval: While architectural research continues to expand maximum context lengths [7] [10], Hong et al. [15] demonstrate that raw capacity does not equate to effective utilization, and Mei et al. [8] recommend context compression and filtering as necessary complements. This study resolves this by prioritizing external retrieval-based augmentation, as context filtering is a mandatory complement to prevent semantic degradation.

Internal vs. External Memory Utility: These are frequently framed as competing strategies, but the literature suggests they are complementary. Internal optimization, such as Gemma 3’s hybrid attention [10], is best suited for single-session reasoning, whereas external retrieval is required for cross-session persistence (as in MemGPT [5] and LONGMEM [4]).

Multi-Hop Efficiency: A conflict exists between iterative retrieval (accu-

rate but slow) and single-step methods. This study looks to the resolution offered by HippoRAG [7], which utilizes graph structures to enable single-step traversal, significantly reducing latency compared to iterative reasoning chains.

Architectural Complexity vs. Resource Constraints: A major tension exists between state-of-the-art performance and deployment feasibility. While systems like MIRIX [1] achieve high accuracy through eight-agent orchestration, they rely on complex function-calling that exceeds the reasoning bandwidth of SLMs. Conversely, ENGRAM [22] demonstrates that a single router and lightweight retriever can match or exceed complex multi-stage systems [2]. This research adopts the latter “simplicity-first” principle as the default for resource-constrained deployment.

2.6.2 Critical Gaps Identified with Evidence

Despite substantial progress, three critical gaps prevent translation of these advances to edge-based personal assistants. The first gap is architectural mismatch between memory system complexity and SLM reasoning capacity. While A-MEM [3] and LONGMEM [4] demonstrate memory augmentation on smaller models, two unresolved questions persist: (1) whether structured text or vector-embedded retrieval better suits SLMs with limited reasoning bandwidth, and (2) whether hybrid-attention architectures like Gemma 3 4B’s 5:1 local-global interleaving can effectively integrate external memory. No study has empirically validated either question under edge deployment constraints.

The second gap is the absence of dynamic disconnected-session evaluation protocols. Existing benchmarks rely on static, pre-recorded logs or synthetic constructions. Real-world personal assistants must retrieve context after hours or days of inactivity, which are scenarios static benchmarks cannot simulate.

The third gap is insufficient optimization guidance for latency-constrained edge deployment. While latency optimizations exist, they assume server-grade compute. No framework optimizes memory design within simultaneous sub-4GB

RAM, <500ms latency, and battery constraints despite on-device feasibility being technically achievable. Section 2.7.5 provides comprehensive analysis of how each study relates to these gaps.

2.6.3 Research Justification and Expected Contributions

The identified gaps collectively justify a systematic evaluation of memory systems for Gemma 3 4B across disconnected sessions under explicit resource constraints. First, it empirically characterizes whether Gemma 3 4B’s hybrid attention architecture can maintain cross-session continuity under external memory augmentation. Unlike the dense-attention Llama models evaluated in A-MEM [3], Gemma 3’s 5:1 local-global interleaving reduces KV-cache memory by 85% [10], but whether this efficiency gain translates to effective memory retrieval across disconnected sessions remains an open empirical question that this study directly answers. Second, it establishes a dynamic disconnected-session evaluation protocol that measures memory retrieval accuracy and task success across temporal gaps (minutes to days) with variable semantic drift, conditions absent from static benchmarks like MSC and LongMemEval [6]. This protocol directly simulates the primary use case of personal edge assistants: resuming context after extended inactivity. Third, it characterizes the accuracy-latency-storage trade-off space for SLM memory systems under edge constraints (sub-4GB RAM, <500ms retrieval latency). By empirically mapping which retrieval strategies achieve >80% task success within these bounds, the study provides the first optimization guidance for deploying memory-augmented SLMs on mobile-class hardware—directly addressing the resource optimization gap left open by Shen et al. [18] and Mem0 [20]. Fourth, it empirically tests Terranova et al.’s [24] hypothesis that smaller foundation models benefit most from simpler RAG-based architectures rather than complex multi-agent memory systems. By comparing lightweight retrieval against autonomous memory management on Gemma 3 4B, the study determines whether the capability-dependency limitation identified by Xu et al. [3]—that memory

quality depends on underlying model capability—holds for hybrid-attention SLMs, establishing whether architectural simplicity is sufficient or whether Gemma 3 4B requires more sophisticated orchestration to achieve cross-session continuity. These contributions address critical knowledge gaps while providing practical deployment guidance for resource-constrained scenarios, which are increasingly relevant as AI systems migrate to on-device deployment.

2.6.4 Synthesis of Related Literature

The reviewed literature encompasses twenty studies spanning 2023 to 2025, collectively advancing understanding across five interconnected research domains: memory architectures for language models, retrieval-augmented generation systems, memory organization strategies, evaluation methodologies, and small language model capabilities. This synthesis organizes these works thematically to identify convergent findings, persistent gaps, and their collective relationship to the present research.

Memory Architectures: Internal Optimization Versus External Augmentation

The literature reveals a fundamental architectural dichotomy between internal context window optimization and external memory augmentation, each suited to different use cases. Chen et al. [11] demonstrated that positional interpolation can extend RoPE-based models from 2K to 32K tokens with less than 2% performance degradation, achieving a $600\times$ smaller interpolation bound compared to extrapolation methods. This work established the technical feasibility of context extension but remained bounded by finite windows and did not address cross-session persistence. The Gemma Team [10] advanced internal optimization further through hybrid attention architecture, employing a 5:1 interleaving of local sliding window and global attention layers that reduces KV-cache memory consumption by approximately 85% while maintaining competitive performance on MMLU,

HumanEval, and MathVista benchmarks. The Gemma 3 4B model achieves performance competitive with models 6–7 $\times$ larger, demonstrating that architectural innovation can mitigate memory overhead. However, evaluation remained confined to single-session static benchmarks, leaving the interaction between hybrid attention and external memory injection uncharacterized—a gap directly addressed by the present study.

External memory augmentation emerged as the solution for persistent cross-session continuity. Packer et al. [5] introduced MemGPT, an OS-inspired hierarchical memory system that pages information between fast in-context storage and slow external storage through function calls, successfully maintaining coherence across multiple sessions. The study’s critical limitation—brittleness when evaluated with GPT-3.5 versus GPT-4—revealed that complex autonomous memory management may exceed the reasoning bandwidth of smaller models, a finding central to the present study’s architectural decisions. Wang et al. [4] proposed LONGMEM, which augments a frozen backbone LLM with a trainable SideNet (approximately 37% of backbone parameters) for chunk-level memory retrieval, achieving perplexity reductions of 1.38–1.62 and 40.5% accuracy on the ChapterBreak benchmark compared to a $\sim$ 28% baseline. While LONGMEM validated memory augmentation on the relatively small GPT-2 (407M parameters), its evaluation was limited to continuous long-form text sequences such as books, not the disconnected multi-session dialogues characteristic of personal assistants. Moreover, the retrieval component introduced computational overhead equivalent to approximately 55% of backbone inference time, establishing the latency constraints that inform the present study’s sub-500ms target. These studies collectively demonstrate that external memory enables cross-session persistence but leave unaddressed how such systems perform on hybrid-attention SLMs under edge deployment constraints.

Retrieval-Augmented Generation: Evolution and Production Deployment

The RAG paradigm has evolved from simple retrieve-read pipelines to sophisticated modular systems capable of iterative reasoning. Gao et al. [14] provided a comprehensive taxonomy organizing RAG into three generational stages: Naive RAG (straightforward retrieval followed by generation), Advanced RAG (incorporating pre- and post-retrieval optimization such as query rewriting and re-ranking), and Modular RAG (flexible orchestration of specialized components including Search, Memory, and Routing modules). Critically, their survey identified inverse scaling laws as an open research question, suggesting that smaller models may effectively compete with larger ones in RAG setups—a hypothesis the present study tests empirically. Gutiérrez et al. [7] demonstrated the viability of specialized RAG through HippoRAG, which employs schemaless knowledge graphs and Personalized PageRank for single-step multi-hop retrieval, achieving 89.5% Recall@5 compared to ColBERTv2’s 68.2% while operating 10–30 $\times$ cheaper and 6–13 $\times$ faster than iterative methods such as IRCoT. Importantly, HippoRAG was evaluated on both closed frontier models (GPT-3.5) and smaller models (Llama-3.1-8B), confirming viability on architectures below the typical ≥ 13 B threshold. However, the study did not evaluate models under 5B parameters or consider edge deployment constraints, gaps the present research addresses.

Production deployment viability hinges critically on latency optimization. Shen et al. [18] systematically decomposed RAG inference latency, finding that with aggressive re-retrieval, RAG components account for 97% of total latency—specifically, 36% from retrieval and 45% from prefill—with Time-To-First-Token increasing from approximately 495ms to 965ms when RAG is added. This overhead directly informed the present study’s sub-500ms retrieval latency target as the threshold for non-disruptive conversational interactions. Chhikara et al. [20] demonstrated that structured memory retrieval can simultaneously improve ac-

curacy and reduce computational overhead, with Mem0 achieving 26% relative improvement over OpenAI’s full-context method on the LOCOMO benchmark while attaining 91% lower p95 latency and over 90% token cost reduction. These efficiency gains establish that external memory is superior to context window expansion for resource-constrained deployment, yet Mem0’s evaluation assumed server-grade computational availability, leaving mobile-class hardware optimization unaddressed. Gan et al. [19] surveyed 582 papers to establish a unified RAG evaluation taxonomy covering retrieval quality, generation quality, safety, and efficiency, identifying a critical gap: the lack of standardized benchmarks for memory persistence across disconnected sessions, as most evaluations focus on continuous contexts or single queries. This gap directly motivates the present study’s dynamic disconnected-session protocol.

Memory Organizations: From Type-Based Partitioning to Cognitive Models

The structured text versus vector-embedded distinction represents a fundamental architectural choice with unresolved implications for SLMs. MemGPT [5] and RMM [21] demonstrate structured text approaches through edit buffers and topic summaries, offering interpretability but requiring explicit reasoning for memory management. In contrast, LONGMEM [4] and Mem0 [20] prioritize retrieval-centric architectures, utilizing cached embeddings and dense vector search to achieve robust semantic matching, often at the cost of embedding overhead. HippoRAG [7] further extends this by integrating knowledge graphs to model associative links, addressing the opacity of flat vector retrieval. While Teranova et al. [24] suggest simpler architectures benefit smaller models, no study distinguishes text-based simplicity from embedding-based simplicity for hybrid-attention SLMs under edge constraints.

Memory organization strategies determine how effectively relevant information can be surfaced during retrieval. Wang and Chen [1] introduced MIRIX, a six-

agent architecture organizing memory into Core, Episodic, Semantic, Procedural, Resource, and Knowledge Vault components, each managed by a dedicated agent. On the LOCOMO benchmark, MIRIX achieved state-of-the-art accuracy of 85.4%, substantially outperforming standard RAG (51.6%) and competing systems such as Zep (79.1%). However, the architecture’s reliance on eight distinct agents and models with strong function-calling capabilities introduces significant orchestration complexity that may overwhelm smaller models—a limitation acknowledged in the study and central to the present research’s investigation of architectural simplicity. Tan et al. [2] pursued a different organizational philosophy with RMM, combining prospective topic-based summarization with retrospective RL-refined retrieval using unsupervised LLM attribution signals as self-supervised rewards. Evaluated on LongMemEval [6] and MSC datasets, RMM achieved up to 73.8% accuracy compared to standard RAG baselines at approximately 64%, demonstrating that principled organization improves multi-session consistency. The system’s acknowledged limitations—RL training overhead and dependency on generator feedback quality—highlight the trade-offs between performance gains and computational cost.

Dynamic knowledge networks represent an alternative to fixed organizational schemas. Xu et al. [3] proposed A-MEM, which draws on Zettelkasten principles to create atomic notes with autonomous linking and memory evolution triggered by new experiences. Of particular significance to the present research, A-MEM was explicitly evaluated on small language models: Llama 3.2 1B and Qwen 2.5 1.5B equipped with A-MEM significantly outperformed baselines such as MemGPT and MemoryBank, with Llama 3.2 1B achieving a ROUGE-L score of 60.23 on adversarial tasks compared to MemGPT’s 50.31. Scalability tests demonstrated exceptional retrieval efficiency, with latency increasing only from 0.31 μ s to 3.70 μ s as memory repositories scaled from 1,000 to 1,000,000 entries. However, the authors acknowledged a critical limitation: “the quality of memory evolution is influenced by the inherent capabilities of the underlying language models,” meaning

smaller models may generate lower-quality contextual descriptions. This unresolved limitation—tested only on dense-attention Llama models—represents a gap the present study characterizes empirically for Gemma 3 4B’s hybrid attention architecture.

Patel and Patel [22] challenged the assumption that complexity is necessary, proposing ENGRAM with the explicit objective of demonstrating that architectural simplicity can match or exceed complex multi-stage systems. ENGRAM organizes conversation into three canonical memory types—episodic, semantic, and procedural—through a single router and retriever, converting each user turn into typed memory records with normalized schemas and dense embeddings. At query time, the system retrieves top-k dense neighbors per type and merges results with simple set operations. Evaluated on LoCoMo and LongMemEval [6], ENGRAM achieved state-of-the-art performance without knowledge graphs, multi-stage retrieval pipelines, or OS-style schedulers, directly supporting the present study’s hypothesis that simpler architectures may suffice for SLMs. However, ENGRAM was not evaluated on hybrid-attention models or under edge deployment constraints, leaving its viability for the present study’s target scenario empirically unvalidated.

Cognitively-inspired approaches offer a complementary organizational principle. Zhong et al. [12] proposed MemoryBank, grounding memory management in the Ebbinghaus Forgetting Curve—a psychological principle describing how memory retention decays exponentially over time unless reinforced through recall. The system translates this into a dynamic updating mechanism where memories are selectively forgotten or strengthened based on elapsed time and recall frequency, with results demonstrating more natural, personalized long-term interactions compared to static storage. MemoryBank’s contribution to the present study is conceptual rather than empirical: its demonstration that selective forgetting is not a failure mode but a desirable design property is particularly relevant for edge deployments where storage capacity is limited. However, the study relied

on qualitative evaluation through the SiliconFriend chatbot without quantitative benchmarking, limiting its empirical generalizability. Mei et al. [8] formalized the broader discipline of Context Engineering through a survey of over 1,400 papers, establishing a unified taxonomy covering Retrieval/Generation, Processing, and Management. Their analysis revealed that while Context Processing innovations address the $O(n^2)$ computational complexity of Transformers, they do not solve the “lost-in-the-middle” phenomenon where performance degrades by up to 73% in long contexts. Consequently, they emphasize Context Management, specifically Memory Hierarchies, as the critical solution for persistent interaction, providing the theoretical framework within which the present study’s memory system operates.

Evaluation Methodologies: Benchmarks and the Context Rot Phenomenon

Rigorous evaluation of memory systems requires benchmarks that capture the complexity of real-world long-term interactions. Wu et al. [6] introduced LongMemEval with the objective of providing a comprehensive benchmark specifically designed to evaluate long-term memory capabilities across five core abilities: information extraction, multi-session reasoning, temporal reasoning, knowledge updates, and abstention. With 500 meticulously curated questions embedded within freely scalable user-assistant chat histories, LongMemEval presents a significant challenge, with commercial chat assistants and long-context LLMs showing 30–60% accuracy drops under extended interaction conditions. The benchmark represents a significant advance and has been accepted at ICLR 2025, establishing it as an evaluation standard. However, despite this advancement, LongMemEval uses synthetic conversations constructed from fixed interaction templates and does not simulate dynamic disconnection with variable temporal gaps—the primary use case for personal edge assistants. Maharana et al. [21] contributed the LOCOMO benchmark for long-term conversational memory, which has become a standard

evaluation instrument used by MIRIX [1], Mem0 [20], A-MEM [3], and ENGRAM [22]. However, like LongMemEval, LOCOMO relies on pre-recorded conversation logs with context lengths around 9K tokens, failing to capture the temporal discontinuities and semantic drift that occur when assistants must retrieve context after hours, days, or weeks of inactivity.

The most critical empirical challenge to context window expansion comes from Hong et al. [15], who conducted a rigorous analysis of long-context model behavior examining 18 models including GPT-4.1, Claude 4, Gemini 2.5, and Qwen3. Using an extended Needle-in-a-Haystack protocol enriched with semantic variations as the evaluation instrument, their study revealed three principal findings: (1) all evaluated models exhibit non-uniform performance degradation as input length increases; (2) needle-question semantic similarity is a critical factor, with performance degrading significantly faster when the needle is semantically ambiguous relative to the query; and (3) the presence of even four distractors causes notable hallucinations, particularly in GPT models, while Claude models tended to abstain rather than confabulate. Contrary to the standard “Lost in the Middle” hypothesis, they found no notable performance variation based on needle position for retrieval tasks; instead, degradation is driven by semantic ambiguity and haystack structure, with coherently structured haystacks performing worse than shuffled ones. These findings provide compelling empirical justification for retrieval-based memory over context window expansion, as raw capacity does not equate to effective utilization. The present study builds on this foundation by pairing Gemma 3 4B’s hybrid attention (which mitigates but does not eliminate Context Rot) with external retrieval to achieve persistent cross-session continuity.

Small Language Models: Capabilities and the Data Quality Principle

The feasibility of deploying memory-augmented language models on edge devices depends critically on whether SLMs can serve as capable reasoning engines

within strict resource budgets. Microsoft [9] introduced Phi-3-mini (3.8 billion parameters) with the explicit objective of challenging conventional scaling assumptions by demonstrating that data quality, rather than parameter count, is the primary determinant of SLM capability. Trained on 3.3 trillion tokens of heavily filtered web data and synthetic data curated to resemble textbook-quality content, Phi-3-mini achieves 68.8% on MMLU and 82.5% on GSM8K, competitive with GPT-3.5’s scores of 71.4% and 78.1%, respectively. On-device feasibility was validated by running the model on an iPhone 14, demonstrating that SLMs can operate within mobile-class hardware constraints. However, the study did not evaluate external memory systems or cross-session persistence, focusing instead on static benchmarks such as RULER for long- context capabilities. This limitation is significant: the data quality innovations that enable Phi-3’s strong single-session performance have not been extended to memory-augmented multi-session scenarios, representing a gap the present study addresses by pairing Gemma 3 4B with an external long-term memory framework.

Terranova et al. [24] directly investigated the question of how memory architecture complexity should scale with model capability through a systematic evaluation of memory-augmented methods on long-context dialogues requiring diverse reasoning strategies. Their study examined full-context prompting, semantic memory through both RAG and agentic memory (A-MEM [3]), episodic memory through in-context learning, and procedural memory through prompt optimization, evaluated consistently across multiple LLMs of varying capability using the LoCoMo benchmark. The key finding is directly relevant to the present study: memory-augmented approaches reduce token usage by over 90% while maintaining competitive accuracy, and critically, memory architecture complexity should scale with model capability—specifically, smaller foundation models benefit most from RAG, while stronger instruction-tuned models can additionally leverage more complex episodic and agentic semantic memory. The authors further find that episodic memory helps LLMs recognize the limits of their own knowledge, mit-

igating metacognitive limitations. This result provides empirical grounding for the present study’s hypothesis that a relatively simple retrieval-based memory framework may be the optimal design for Gemma 3 4B, rather than the more complex autonomous architectures validated primarily on larger models. However, Terranova et al.’s validation was conducted on dense-attention models, and the hypothesis remains untested on hybrid-attention SLMs under dynamic disconnected sessions and edge constraints—which is precisely the empirical question the present study answers.

Convergence, Contradictions, and Research Justification

Analysis of the reviewed literature reveals five strong points of convergence. First, external memory systems consistently improve performance over non-augmented baselines across all evaluated frameworks: LONGMEM [4] demonstrates perplexity reductions of 1.38–1.62, RMM [2] achieves more than 5% accuracy improvement on LongMemEval [6], HippoRAG [7] surpasses ColBERTv2 by 16–20 percentage points on multi-hop QA, and MIRIX [1] achieves 33% higher accuracy than standard RAG on LOCOMO [21]. Second, memory organization matters beyond raw storage capacity: RMM’s topic-based approach [2], MIRIX’s type-based architecture [1], A-MEM’s dynamic linking [3], and ENGRAM’s lightweight typed retrieval [22] all outperform flat vector retrieval, indicating that how information is structured is as consequential as how much is stored. Third, sub-500ms retrieval latency is achievable through index optimization, as demonstrated by Shen et al. [18] and further confirmed by Mem0’s 91% latency reduction [20], though total generation latency with RAG remains higher than without. Fourth, context utilization consistently falls below advertised context capacity: Hong et al. [15] demonstrate that effective accuracy degrades significantly in extended contexts through the Context Rot phenomenon, validating the necessity of retrieval-based augmentation over context window expansion. Fifth, data quality enables competitive small model performance, as Phi-3 [9] demonstrates by rivaling models approximately

50 $\times$ larger in parameter count.

The literature also presents several productive tensions that resolve upon closer examination. The first tension is between context window growth and context rot: while architectural research continues to expand maximum context lengths (Chen et al. [11], Gemma 3 [10]), Hong et al. [15] demonstrate that raw capacity does not equate to effective utilization. These are complementary rather than contradictory findings: internal context optimization is suited for single-session reasoning, while external retrieval is necessary for cross-session persistence. The second tension is between iterative and single-step multi-hop retrieval: HippoRAG’s [7] graph structure resolves this by enabling single-step traversal for multi-hop questions, reducing latency by an order of magnitude relative to iterative methods. The third tension is between architectural complexity and performance: MIRIX [1] achieves state-of-the-art performance through eight-agent orchestration, while ENGRAM [22] matches or exceeds comparable systems through a single router and retriever. This suggests that architectural simplicity should be the default assumption for resource-constrained deployment, with complexity added only when empirically justified—a principle guiding the present study’s design.

Despite this substantial body of work, three critical gaps prevent the translation of these advances to edge-based personal assistants. The first gap is architectural mismatch between memory system complexity and SLM reasoning capacity. While A-MEM [3] and LONGMEM [4] demonstrate that memory augmentation can work on smaller models, the dominant production-grade architectures—MemGPT [5] and MIRIX [1]—require autonomous function-calling and multi-agent orchestration that may exceed the reliable reasoning bandwidth of models under 5B parameters. Although ENGRAM [22] demonstrates that lightweight typed memory can match complex architectures on standard benchmarks, it has not been evaluated specifically on hybrid-attention SLMs like Gemma 3 4B. Furthermore, while Terranova et al. [24] confirm that smaller foundation models ben-

efit most from simpler RAG-based architectures, this recommendation has not yet been validated with controlled empirical evidence across dynamic disconnected sessions. Critically, no study has empirically validated memory system performance on hybrid-attention SLMs under 5B parameters. Gemma 3 4B’s 5:1 local-global attention interleaving represents a fundamentally different architecture from the dense-attention Llama models evaluated in A-MEM [3] and LONGMEM [4], yet its interaction with external memory retrieval remains uncharacterized.

The second gap is the absence of dynamic disconnected-session evaluation protocols. Existing benchmarks for multi-session consistency—MSC used by MemGPT [5] and LongMemEval [6] used by RMM [2]—rely on static, pre-recorded conversation logs or synthetic dialogue constructions. Real-world personal assistants must retrieve context after hours, days, or weeks of inactivity, scenarios that static benchmarks cannot simulate. No existing evaluation protocol measures memory retrieval accuracy under dynamic disconnection, where session gaps introduce semantic drift and temporal ambiguity beyond what continuous dialogue benchmarks capture. This is not merely a methodological refinement but a fundamental mismatch between evaluation instruments and deployment reality: the primary use case for personal edge assistants is resuming context after extended inactivity, yet no standardized protocol exists to evaluate this capability.

The third gap is insufficient optimization guidance for latency-constrained edge deployment. While Shen et al. [18] quantify RAG latency overhead and Mem0 [20] achieves 91% p95 latency reduction, these optimizations assume server-grade compute with unlimited memory budgets. Edge deployment imposes strict constraints: sub-4GB RAM availability, battery-limited inference cycles, and user-perceptible latency thresholds under 500ms. No framework currently optimizes memory system design within these simultaneous constraints, despite the fact that on-device feasibility demonstrated by Phi-3 [9] is now technically achievable for SLMs. The absence of optimization frameworks for the accuracy-latency-memory Pareto frontier under edge deployment constraints leaves practitioners

without guidance on which retrieval strategies achieve acceptable task success within mobile-class hardware bounds.

These gaps collectively justify a systematic evaluation of memory systems for Gemma 3 4B across disconnected sessions under explicit resource constraints. By evaluating efficient decoupled retrieval rather than autonomous OS-style memory management under a dynamic disconnected-session protocol, the present study provides the first controlled empirical evidence of whether hybrid-attention SLMs can maintain task-relevant memory across temporal gaps within edge deployment constraints. The study empirically characterizes whether Gemma 3 4B’s hybrid attention architecture can maintain cross-session continuity under external memory augmentation, a question left unanswered by prior work on dense-attention models. It establishes a dynamic disconnected-session evaluation protocol that measures memory retrieval accuracy and task success across temporal gaps (minutes to days) with variable semantic drift, directly simulating the primary use case of personal edge assistants. It characterizes the accuracy-latency-storage trade-off space for SLM memory systems under edge constraints (sub-4GB RAM, <500ms retrieval latency), providing the first optimization guidance for deploying memory-augmented SLMs on mobile-class hardware. Finally, it empirically tests Terranova et al.’s [24] hypothesis that smaller foundation models benefit most from simpler RAG-based architectures rather than complex multi-agent memory systems, determining whether the capability-dependency limitation identified by Xu et al. [3] holds for hybrid-attention SLMs. These contributions address critical knowledge gaps while providing practical deployment guidance for resource-constrained scenarios, which are increasingly relevant as AI systems migrate to on-device deployment. The findings directly inform the feasibility boundaries of on-device personal assistants with long-term continuity.

2.7 Summary

This chapter reviewed the state of the art in retrieval-augmented generation and memory systems, synthesizing findings across memory architectures, vector-embedded retrieval, memory organization strategies, evaluation methodologies, and small language model capabilities. The literature demonstrates strong convergent evidence that external memory systems significantly enhance model performance, that memory organization matters beyond raw storage, and that sub-500ms retrieval latency is achievable. The Context Rot findings provide compelling empirical justification for retrieval-based memory over context window expansion.

Despite these advances, three critical gaps stifle deployment of capable memory-augmented assistants on edge devices: architectural mismatch between complex memory frameworks and SLM reasoning capacity, absence of dynamic disconnected-session evaluation protocols, and insufficient optimization guidance for latency-constrained edge deployment. Section 2.7.5 provides comprehensive synthesis demonstrating how each reviewed study relates to these gaps and the present research.

These gaps justify the proposed research: a systematic evaluation of a Long-Term Memory Framework for Gemma 3 4B. By shifting focus from autonomous OS-style memory management to efficient decoupled retrieval, and evaluating under a novel dynamic disconnected-session protocol, this thesis aims to provide the first empirical evidence of how hybrid-attention SLMs can achieve persistent personalization in resource- constrained environments.

Figure 2.1 presents the conceptual framework guiding this research, structured as a four-layer progression from fundamental challenges to the present study’s contributions:

Layer 1 identifies the three core challenges motivating this work: the Context Rot phenomenon, which demonstrates non-uniform performance degradation in extended contexts [15]; resource constraints inherent to edge AI deployment,

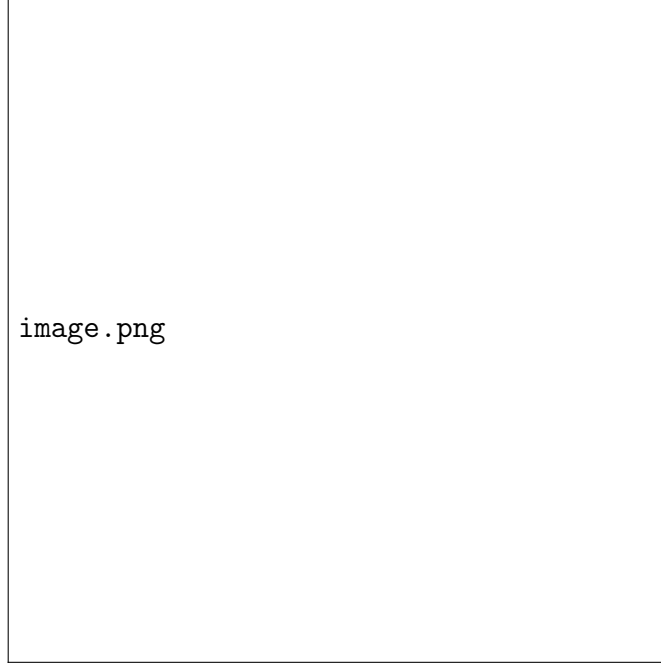


Figure 2.1: Conceptual Framework

including sub-4GB RAM budgets and sub-500ms latency thresholds; and the cross-session continuity gap resulting from LLMs’ lack of persistent memory across temporal discontinuities.

Layer 2 synthesizes the solution approaches identified in the literature review, encompassing memory architectures (internal context optimization versus external retrieval systems), RAG system evolution from naive to modular paradigms, memory organization strategies ranging from type-based to cognitively-inspired approaches, and emerging SLM capabilities demonstrating competitive performance through data quality optimization.

Layer 3 articulates the three critical research gaps that existing literature leaves unaddressed: the architectural mismatch between complex memory frameworks and SLM reasoning capacity, particularly for hybrid-attention models like Gemma 3 4B; the absence of dynamic disconnected-session evaluation protocols that simulate real-world temporal gaps and semantic drift; and the lack of optimization frameworks for the accuracy-latency-memory Pareto frontier under edge deployment constraints.

Layer 4 positions the present study as directly addressing these gaps through

four specific contributions: empirical characterization of Gemma 3 4B’s interaction with external memory retrieval, establishment of a dynamic disconnected-session protocol, mapping of resource-constrained trade-off spaces, and hypothesis testing of whether simpler RAG architectures suffice for SLMs.

This framework establishes the theoretical foundation for investigating whether hybrid-attention small language models can achieve persistent personalization within the strict resource boundaries of on-device deployment.

Chapter 3:

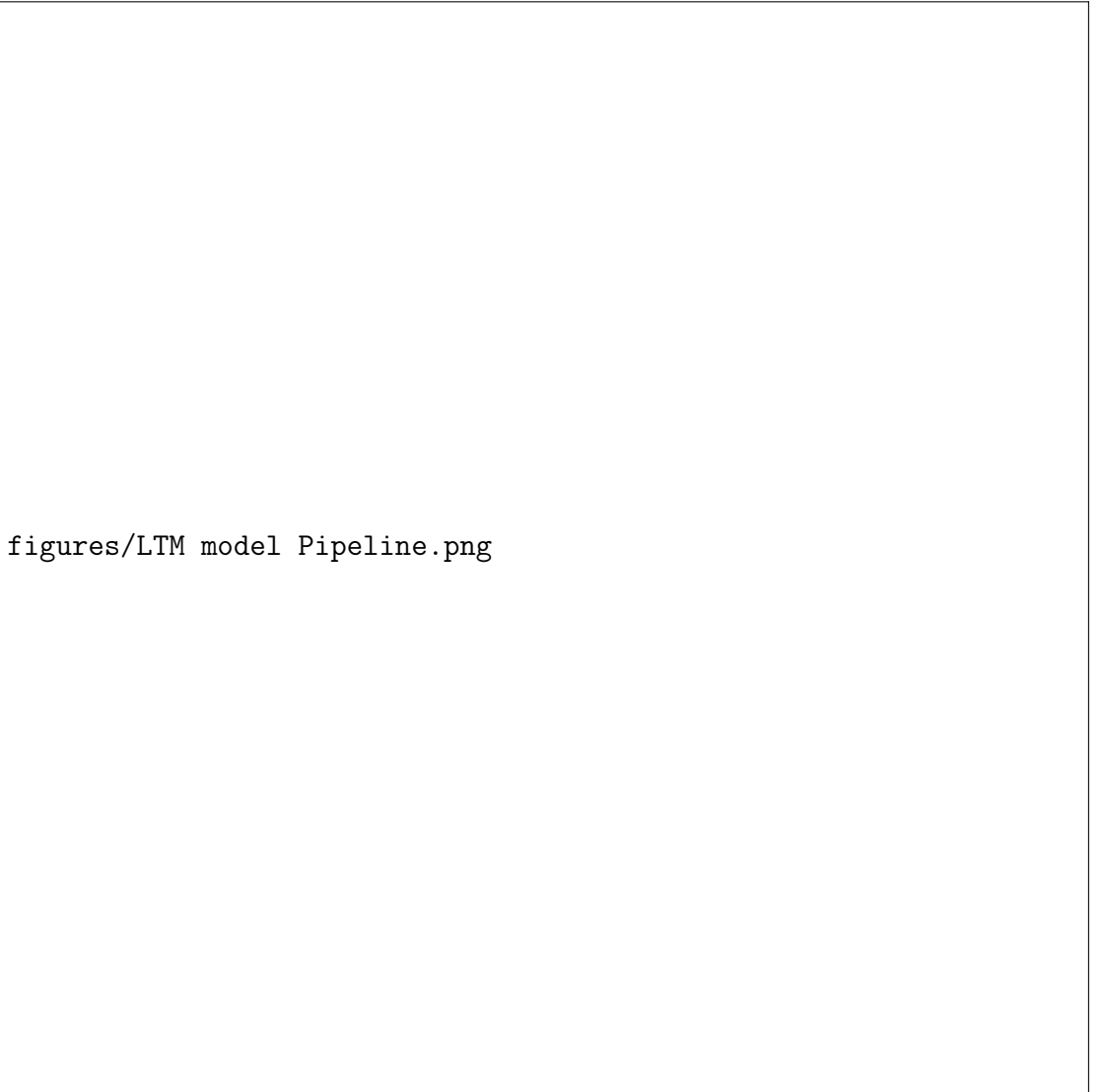
METHODOLOGY

This chapter discusses the advancement of localized artificial intelligence—a transition from the traditionally stateless inference cycles of Small Language Models (SLMs) toward stateful, memory-augmented architectures. While Large Language Models (LLMs) often rely on vast context windows or immense parametric knowledge to maintain continuity, SLMs are constrained by limited parameter counts and the necessity of operating on consumer-grade hardware [10]. The study is structured as an experimental-comparative design evaluating three distinct memory conditions against two purpose-built long-term memory benchmarks. The first condition is the stateless baseline: Gemma 3 4B operating with no external memory store, receiving only the current query at inference time. The second condition is BM25 lexical retrieval: Gemma 3 4B augmented with a SQLite FTS5 inverted index, retrieving the top-5 most relevant memory records via probabilistic term-frequency scoring. The third condition is dense semantic retrieval: Gemma 3 4B augmented with a FAISS IndexFlatL2 index populated by EmbeddingGemma 300M vectors, retrieving the top-5 most semantically similar memory records. All three conditions are evaluated on LongMemEval and LoCoMo — two peer-reviewed benchmarks natively designed for multi-session memory evaluation — eliminating the need for artificial session-break adaptation of general-purpose datasets. The approach adopted is an empirical research design centered on controlled experimental evaluation and quantitative data analysis.

3.1 Research Design

The investigative framework is grounded in a positivist worldview, emphasizing objective measurement and the establishment of causal relationships between

memory architecture variables and model performance outcomes. This orientation facilitates a deductive research approach, where the effectiveness of memory-augmented architectures is tested against established benchmarks of task success and recall accuracy. The study is structured as an experimental-comparative design that progresses through distinct phases of model selection, architectural integration, scenario-based testing, and statistical validation.



figures/LTM model Pipeline.png

Figure 3.1: Model Pipeline

The operational pipeline follows a two-phase structure for each dataset. In the ingestion phase, the conversation history provided by each benchmark is fed into the memory backend session-by-session, simulating a system that has accumulated memory over the course of real interactions. In the evaluation phase,

each benchmark question is passed through the retrieve-inject-generate pipeline: the query triggers memory retrieval, the top-5 retrieved records are injected into the model context alongside the query, and Gemma 3 4B generates a response. All five core metrics are logged programmatically at each inference step. For the baseline condition, no ingestion phase occurs and no memories are injected.

The baseline condition consists of Gemma 3 4B operating with no external memory store and no session history injected into the context. For each scenario, the model receives only the current session’s prompt, meaning it has no access to information from prior sessions. When a scenario requires recall of information from a previous session, the model must rely solely on its parametric knowledge, which by design does not contain session-specific user data. Responses in the baseline condition are evaluated using the same automated metrics as the experimental conditions: a response that fails to recall required prior information is scored as a task failure, producing a 0 for Task Success Rate. This baseline directly measures the performance cost of the stateless condition and provides the lower-bound reference against which both memory architectures are compared.

3.2 Technical Specifications

The selection of the primary language model represents a critical decision in the experimental setup, as the base model’s inherent reasoning capacity and architectural efficiency determine the efficacy of external memory integration. The Gemma 3 4B model is identified as the primary candidate, with comparisons drawn to the Phi-3-mini 4K and Llama 3.2 3B architectures to justify its selection as the experimental core.

3.2.1 Primary Model Architecture: Gemma 3 4B

Gemma 3 4B is a 4-billion-parameter multimodal open model built upon the research foundations of the Gemini family [10]. Its selection is predicated on several architecturally significant capabilities [10], including a native 128K context

window and a highly efficient hybrid attention mechanism. The model’s architecture incorporates a 5:1 interleaving of local sliding window self-attention and global self-attention layers [10]. The researchers define the computation cost $T(N)$ for a sequence of length N as:

$$T(N) \approx L_{\text{local}} \cdot (N \cdot W) + L_{\text{global}} \cdot N^2 \quad (3.1)$$

where W is the sliding window size, L_{local} is the number of local layers, and L_{global} is the number of global layers.

The local attention layers operate with a constrained sliding window of 1,024 tokens, ensuring that each token only attends to its immediate neighbors [10]. In contrast, global attention layers, occurring every sixth layer, maintain access to the full sequence [10]. This arrangement substantially reduces attention computations and reduces KV-cache memory requirements from approximately 60% of total memory to less than 15%—an 85% reduction that is vital for resource-constrained local deployments [10]. Furthermore, Gemma 3 4B adopts a QK-norm operation to replace the soft-capping mechanism found in previous versions, enhancing numerical stability and scaling performance during long-context reasoning [10].

Parameter	Specification	Rationale for Selection
Model Size	4 Billion Parameters	Optimal balance between reasoning density and RAM overhead.
Context Window	128,000 Tokens	The high ceiling allows for controlled window testing (4K).
Attention Scheme	5:1 Local-Global Hybrid	Minimizes KV cache explosion; isolates external memory effects.
Training	4 Trillion Tokens	Reasoning-dense data (web, math, code) ensures base utility.
MMLU Score	59.6% (Pre-trained)	Competitive performance baseline for an SLM of its scale.
Hardware Fit	16GB RAM Laptop	Matches targeted deployment constraints for real-world use.

Table 3.1: Gemma 3 4B specifications

3.2.2 Comparative Model Evaluation

To validate the choice of Gemma 3 4B, alternative SLMs were assessed against the experimental requirements. The Phi-3-mini 4K serves as a backup candidate, utilizing a “data optimal” training regime with high-quality synthetic data to achieve 69% MMLU despite its small parameter count [9]. However, the standard attention mechanism of Phi-3-mini lacks the memory-efficient hybrid architecture of Gemma 3, which is central to investigating external memory overhead [10]. Similarly, while Llama 3.2 3B offers fast inference, its smaller parameter count may limit complex reasoning in memory-augmented tasks [26].

3.3 External Memory System

The methodology implements and compares two distinct memory paradigms: a structured relational system and a vector-embedded semantic system. These systems transition the SLM from a purely parametric memory state to a non-parametric continual learning state [27].

3.3.1 (Structured Text Memory:) BM25 via SQLite FTS5 Lexical Retrieval

The Structured Text Memory condition evaluates lexical, keyword-based retrieval implemented as a sparse retrieval system. Rather than relying on dense neural representations, it stores historical interactions as plain text records and retrieves them through exact and near-exact vocabulary matching between the query and the stored documents, prioritizing transparency and low computational overhead.

Storage Backend: SQLite FTS5

The physical storage and indexing layer for this memory condition is **SQLite’s Full-Text Search extension, FTS5** [28]. FTS5 is a built-in SQLite module that maintains an *inverted index*—a data structure that maps every distinct vocabulary token to the set of rows in which that token appears, together with positional metadata. This inverted index is the structural foundation on which the BM25 ranking function operates.

SQLite FTS5 is selected over alternative sparse-retrieval backends (e.g., standalone Elasticsearch or OpenSearch) for three deployment-critical reasons. First, FTS5 is embedded within the same process as the inference pipeline: it requires no separate server, no network socket, and no additional system service, producing a zero-overhead deployment footprint consistent with the edge-device constraints of this study. Second, FTS5 persists its inverted index as a standard SQLite database file, meaning that the memory store survives process termination and can be resumed across sessions without re-indexing, a prerequisite for genuine long-term memory. Third, FTS5 exposes a native BM25 ranking function through SQL’s `bm25()` auxiliary function, removing the need for a separate scoring library and simplifying the retrieval pipeline.

Each memory entry is inserted into the FTS5 virtual table as a single row containing the following fields: a `session_id` identifying the originating session, a `timestamp` recording the absolute time of the interaction, and a `content` field holding the tokenized interaction text. The FTS5 tokenizer applies Unicode-aware case-folding and punctuation stripping before building the inverted index, ensuring that surface-form variants such as “User” and “user” are treated as the same token without requiring explicit pre-processing.

Retrieval Mechanism via BM25 Ranking

Retrieval is governed by the BM25 (Best Match 25) algorithm [23], the established standard for sparse information retrieval. Given a query q composed of terms $q_1, q_2, \dots, q_n$ and a corpus of N stored documents, BM25 scores each document d as:

$$\text{BM25}(d, q) = \sum_{i=1}^n \text{IDF}(q_i) \cdot \frac{f(q_i, d) \cdot (k_1 + 1)}{f(q_i, d) + k_1 \cdot \left(1 - b + b \cdot \frac{|d|}{\text{avgdl}}\right)} \quad (3.2)$$

where $f(q_i, d)$ is the raw term frequency of token q_i in document d , $|d|$ is the document length in tokens, avgdl is the mean document length across the corpus, and k_1 and b are saturation and length-normalization hyperparameters. IDF is computed as $\ln\left(\frac{N - n(q_i) + 0.5}{n(q_i) + 0.5} + 1\right)$, where $n(q_i)$ is the number of documents containing q_i . The implementation uses FTS5's default hyperparameters ($k_1 = 1.2$, $b = 0.75$), which have been empirically validated across a wide range of text retrieval benchmarks [23].

The retrieval query is issued as a single SQL statement of the form:

```
SELECT content, session_id, timestamp,
       bm25(memory_fts) AS score
FROM memory_fts
WHERE memory_fts MATCH :expanded_query
ORDER BY score
LIMIT :top_k;
```

Because FTS5's `bm25()` returns negative values (lower magnitude = higher relevance in SQLite's implementation), ordering by ascending score surfaces the top-ranked documents. The top $k = 5$ results are retrieved for each query, consistent with the $k = 5$ setting applied to the vector-embedded condition.

Query Expansion

To mitigate the strict exact-match limitation of sparse retrieval—where a query containing “medication” will not retrieve a document containing only “drug” without synonym coverage—the structured memory pipeline incorporates a **query expansion** step prior to issuing the FTS5 search.

Gemma 3 4B performs this expansion via a constrained system prompt that instructs it to generate up to five synonyms and lemmatized keyword variants for the user query. The expanded terms are concatenated into an FTS5 `MATCH` expression using the `OR` operator, broadening the effective vocabulary coverage while retaining the BM25 ranking function as the relevance arbiter.

Using the same model for both expansion and generation is an intentional design decision: it ensures that the sparse memory pipeline remains entirely self-contained within the Gemma 3 4B architecture, preserving experimental isolation of the memory architecture as the independent variable.

Recency Weighting

Standard BM25 is atemporal: it assigns identical scores to a document stored one minute ago and one stored one month ago, given identical term statistics. For a long-term memory system where recent interactions are typically more relevant than older ones, this is a practical limitation. To address this, the pipeline applies a **recency weight** as a post-retrieval re-ranking step. Each candidate document’s raw BM25 score is adjusted by a time-decay factor:

$$\text{score}_{\text{final}}(d) = \text{BM25}(d, q) \cdot e^{-\lambda \cdot \Delta t(d)} \quad (3.3)$$

where $\Delta t(d)$ is the elapsed time in hours since document d was stored, and λ is a decay constant set to 0.01 (corresponding to a half-life of approximately 69 hours). This formulation mirrors the Ebbinghaus Forgetting Curve principle adopted by MemoryBank [12], translating cognitive memory theory into a practical retrieval

heuristic. The top- k documents are selected from the recency-adjusted ranking.

3.3.2 Vector-Embedded Semantic Memory

Architecture 2 focuses on semantic proximity, enabling the model to retrieve memories that may not share exact words with the user query but share underlying meaning [14]. This system relies on EmbeddingGemma (308M), a lightweight 308-million-parameter transformer that maps text to a dense vector space. To optimize for resource-constrained hardware, this architecture leverages EmbeddingGemma’s Matryoshka Representation Learning (MRL), which allows the model to truncate its output vectors from 768 dimensions down to 128 dimensions while maintaining state-of-the-art performance. This truncation drastically reduces the RAM required to store the retrieval index.

The retrieval process involves three primary steps: encoding the user query into a corresponding 128-dimensional vector, performing a similarity search within a FAISS (Facebook AI Similarity Search) index, and injecting the top-K results into the model’s active context window [29]. For the experimental phase of this study, the methodology employs FAISS IndexFlatL2 on normalized vectors, which computes exact Euclidean distances to guarantee optimal retrieval results without approximation error. This choice is highly appropriate for the controlled experimental context; because each session scenario generates a bounded memory store of fewer than 500 entries, IndexFlatL2 remains computationally tractable and achieves sub-500ms retrieval latency on the target hardware. While approximate indices such as HNSW-SQ become advantageous at production scale (thousands to millions of entries), at the experimental scale of this study, the exact-search guarantee of IndexFlatL2 is preferred to eliminate retrieval approximation as a confounding variable.

Feature	Structured Memory (Inverted Index)	Vector Memory (FAISS)
Primary Logic	Lexical Matching and Recency	Semantic Vector Similarity
Search Mechanism	BM25 Ranking	L2 Euclidean / Cosine Similarity
Data Format	Inverted Index	128-Dimensional Vectors
Failure Mode	Lexical Gap (No direct match)	Hallucination (Aggressive matching)
Efficiency	Extremely low CPU overhead	Moderate CPU/Memory overhead

Table 3.2: Differences between Structured Memory and Vector-Embedded Memory

3.4 Scenario Design and Data Engineering

3.4.1 Evaluation Datasets

The evaluation datasets are selected on the basis of three criteria: native multi-session conversation structure requiring no artificial adaptation, existing adoption and validation in the memory systems literature, and direct coverage of the five core metrics defined in this study. Two datasets satisfy these criteria.

Property	LongMemEval [6]	LoCoMo [21]
Purpose	Evaluate long-term memory capabilities of chat assistants	Evaluate long-term conversational memory across extended interactions
Data Type	Multi-session user-assistant conversation histories	Synthetic conversations via human-machine pipeline
Temporal Span	Not specified	Weeks to months
Context Length	Freely scalable	9,000–26,000 tokens
Question Types	Information extraction, multi-session reasoning, temporal reasoning, knowledge updates, abstention	Single-hop, multi-hop, temporal, open-domain, adversarial
Memory Failure Modes Tested	Fact recall failure, cross-session connection failure, temporal misattribution, knowledge update errors, hallucination	Exact retrieval failure, cross-session reasoning failure, temporal misunderstanding, open recall degradation, retrieval confusion
Activation of Study Mechanisms	Tests all five memory conditions across disconnected sessions	Preserves original timestamps; activates BM25 recency weighting
Retrieval Robustness Test	Abstention questions test hallucination boundary	Adversarial questions test resistance to semantically confusing retrievals
Studies That Use This Benchmark	Wu et al. [6], RMM [2]	A-MEM [3], MIRIX [1], ENGRAM [22], Mem0 [20]
Role in This Study	Primary evaluation instrument	Secondary evaluation instrument

Table 3.3: Comparison of Evaluation Datasets

LongMemEval (Wu et al., 2025, accepted ICLR 2025) [6] is a benchmark specifically designed to evaluate long-term memory capabilities of chat assistants. It embeds 500 carefully curated questions within freely scalable multi-session user-assistant conversation histories. Each question targets one of five memory capability types: information extraction, which tests recall of a specific fact stated in a prior session; multi-session reasoning, which requires combining facts from two or more sessions; temporal reasoning, which requires reasoning about when events occurred relative to each other; knowledge updates, which tests whether the system correctly applies a correction made to previously stored information; and abstention, which tests whether the system correctly declines to answer when

the required information was never provided. Wu et al. [6] report that commercial chat assistants and long-context LLMs show a 30–60% performance drop on this benchmark under extended interaction conditions, confirming that it constitutes a genuine challenge to existing memory systems. LongMemEval is selected as the primary evaluation instrument because its five question types directly operationalize the memory failure modes most relevant to a personal assistant deployment: failing to recall facts, failing to connect facts across sessions, misattributing timing, overriding valid information with outdated records, and hallucinating information never stated.

LoCoMo (Maharana et al., 2024, ACL 2024) [21] provides long-term conversational memory data derived from synthetic conversations constructed via a human-machine pipeline spanning weeks to months, with context lengths averaging between 9,000 and 26,000 tokens. Each conversation log is paired with questions of five types: single-hop, which requires retrieving one fact from one session; multi-hop, which requires connecting facts across multiple sessions; temporal, which tests understanding of time-related information; open-domain, which tests general conversational recall; and adversarial, which uses questions specifically designed to confuse the retrieval system through semantic similarity to incorrect answers. LoCoMo is selected as the secondary evaluation instrument because it introduces real temporal distribution across sessions—with original timestamps preserved—which activates and tests the recency weighting mechanism in the BM25 condition, and because its adversarial questions provide the most rigorous test of retrieval robustness for both memory backends. Additionally, LoCoMo is the benchmark used by A-MEM [3], MIRIX [1], ENGRAM [22], and Mem0 [20] in this study’s RRL, making results directly comparable to those systems.

3.4.2 Dataset Ingestion Protocol

!!!! FILL THIS UP !!!!

3.4.3 Metrics Computation

The five core metrics evaluated in this study are summarized in Table 3.4.

Metric	Definition	Measurement Method	Target	Grounded In
Task Success Rate	Proportion of questions answered correctly against ground truth	LLM-as-a-Judge (LongMemEval); per-type scoring (LoCoMo)	>80%	[6], [21]
Recall Accuracy	Token-level F1 score between response and ground-truth answer	Partial string match; F1 scoring	Relative improvement over baseline; reference floor: 40–70% for non-augmented systems [6]	[3], [22], [6]
Memory Error Rate	Proportion of responses containing ungrounded or contradictory claims	Post-inference check against top-5 retrieved records and conversation history	<15%	Section 3.5.1
Retrieval Latency	Wall-clock time from query issued to top-5 results returned	<code>time.perf_counter</code> in milliseconds	500ms	[18]
Context Token Utilization	Total prompt token count as percentage of 128,000-token context window	Programmatic token count at inference	<70%	[15], [10]

Table 3.4: Core evaluation metrics, measurement methods, targets, and empirical grounding.

Task Success Rate and Recall Accuracy measure the correctness of generated responses against ground-truth labels, with Task Success Rate applying an LLM-as-a-Judge protocol for LongMemEval following Wu et al. [6], who explicitly note that exact-match strategies produce inaccurate evaluations for flexible

conversational answers, and per-type scoring for LoCoMo following Maharana et al. [21]. Recall Accuracy is evaluated primarily as a comparative metric across the three memory conditions, with relative improvement over the stateless baseline as the key indicator of memory system effectiveness, consistent with the evaluation methodology adopted by A-MEM [3] and ENGRAM [22] on the LoCoMo benchmark. As a reference point, Wu et al. [6] document that non-augmented systems experience 30–60% performance degradation on LongMemEval under extended interaction conditions, establishing the performance floor against which memory-augmented conditions are interpreted. Memory Error Rate operationalizes retrieval failure by flagging any response claim untraceable to retrieved records or the current query, covering both retrieval misattribution and memory-induced confabulation. Retrieval Latency and Context Token Utilization measure system efficiency: latency is logged at the memory backend level using `time.perf_counter()`, with a sub-500ms target grounded in Shen et al. [18]; token utilization monitors injected context as a percentage of Gemma 3 4B’s 128,000-token window [10], with a <70% threshold derived from the Context Rot phenomenon documented by Hong et al. [15].

3.4.4 Experimental Protocol

Each of the three memory conditions is evaluated independently on the complete question set of both datasets. The experimental procedure is as follows:

Initialization: For each dataset-condition combination, the memory backend is initialized empty. For BM25, the SQLite FTS5 database is created fresh. For dense retrieval, the FAISS IndexFlatL2 is initialized with zero entries. For baseline, no backend is initialized. Ingestion phase: The conversation history for each data point is ingested into the memory backend following the protocol in Section 3.4.1. For the baseline condition, this phase is skipped.

Evaluation phase: For each question, the pipeline generates a response under the active memory condition. The top-k parameter is fixed at k=5 for both

memory conditions.

Metric logging: For each response, Task Success Rate, Recall Accuracy, Memory Error Rate, Retrieval Latency, and Context Token Utilization are logged programmatically to a CSV file with the following fields: `dataset`, `condition`, `question_id`, `question_type`, `response_text`, `ground_truth`, `is_correct`, `f1_score`, `memory_error_flag`, `latency_ms`, `token_count`.

LLM Judge sampling: After all automated metrics are logged, a stratified sample of responses is drawn for LLM-as-Judge evaluation following the protocol in Section 3.5.3.

Reset: Memory backend is cleared between data points as described in Section 3.4.1.

Aggregation: Metrics are aggregated across all questions per dataset, per condition, and per question type, producing mean scores and standard deviations for each combination.

3.5 Performance Evaluation Protocols

The methodology employs five core metrics to quantify the impact of memory architectures on SLM performance. These metrics are grounded in RAG architecture standards and memory system benchmarks.

3.5.1 Task Success and Recall Accuracy

Task Success Rate is defined as the percentage of multi-session tasks completed correctly by the agent. A target of $> 80\%$ is established, reflecting the boundary of real-world viability for autonomous assistants. Cross-Session Recall Accuracy measures the retrieval effectiveness of the memory modules, with a target of $> 80\%$. This accounts for the inherent complexity of retrieving historical facts while maintaining utility across gaps.

Memory Error Rate measures the proportion of scenarios in which the memory system produces incorrect cross-session information, encompassing two failure

modes: retrieval misattribution, in which stored information is retrieved but applied to the wrong context or query; and memory-induced confabulation, in which the model generates session-specific claims absent from the memory store entirely. Both failure modes produce the same practical outcome—incorrect cross-session information delivered to the user and are therefore evaluated under a single metric. This unified treatment is consistent with the operational focus of this study on deployment performance rather than mechanistic failure attribution. A target of $<15\%$ is established, reflecting acceptable error bounds for personal assistant applications and informed by confabulation rates documented in memory-augmented systems under distractor conditions [5][15].

3.5.2 Operational Resource Efficiency

Retrieval Latency is measured as the time required to fetch memories from the database or index. Standards for interactive RAG systems specify a target of $<500\text{ms}$ to maintain natural conversational flow. Furthermore, Context Token Utilization is monitored to ensure that retrieval does not overwhelm the SLM’s active window. The target is to maintain utilization below 70% of the available token budget, a threshold established in the study’s scope as consistent with the operational requirements of edge deployment. This conservative target is justified by the Context Rot phenomenon [15], which demonstrates non-uniform performance degradation as input length approaches and exceeds effective context capacity. Specifically, exceeding 85% utilization has been associated with a 23% degradation in retrieval accuracy; the $<70\%$ target provides a safety margin to prevent this degradation while still enabling meaningful memory injection. Lower utilization ratios thus indicate more efficient and reliable memory retrieval within the model’s effective context boundary.

Metric	Target	Justification
Task Success Rate	> 80%	Threshold for practical utility in agentic AI.
Recall Accuracy	> 75%	Benchmarked against MemoryBank performance [12].
Retrieval Latency	< 500ms	Industry target for real-time responsiveness.
Token Utilization	< 70%	Prevents degradation from context window saturation.
Memory Error Rate	< 15%	Derived from the inverse of success targets.

Table 3.5: Justification for Evaluation Metric Goals

3.5.3 Response Quality Evaluation via LLM-as-Judge

Theoretical Basis: The LLM-as-judge paradigm was formally established by Zheng et al. [30], who conducted a systematic study of using strong LLMs as surrogate evaluators for open-ended chat assistant responses. Introducing MT-Bench, an 80-question benchmark covering eight categories of multi-turn conversational tasks — and Chatbot Arena, a crowdsourced platform collecting over 30,000 human preference votes, Zheng et al. [30] verify that strong LLM judges achieve over 80% agreement with human preferences, the same level of agreement observed between human evaluators themselves, establishing LLM-as-judge as a scalable and explainable approximation of human preference evaluation. Zheng et al. [30] further characterize four known limitations of the paradigm position bias, verbosity bias, self-enhancement bias, and limited reasoning ability and propose mitigation strategies including few-shot judging, chain-of-thought scoring prompts, and reference-guided grading.

For the present study, Gemini 2.5 Flash is selected as the judge model on the basis of its demonstrated capability as an evaluator in recent LLM-as-judge research, its cost efficiency at thesis-scale evaluation, and its accessibility via the Google AI Studio API. Known biases are mitigated through structured rubric-guided scoring prompts with explicit per-dimension criteria, consistent prompt formatting across all evaluated responses, and fixed presentation order of retrieved memories and response text.

Sampling strategy: LLM-as-Judge evaluation is applied to a stratified sam-

ple rather than the full question set, for two reasons: API cost management at thesis scale, and the fact that judge evaluation is most informative on cases where automated metrics are ambiguous — partial credit cases, near-correct responses, and memory error borderline cases. The sampling procedure is as follows: for each dataset $\times$ condition combination, 50 responses are sampled using stratified sampling across question types. For LongMemEval this produces 10 responses per question type (extraction, multi-session, temporal, update, abstention) $\times$ 3 conditions $\times$ 2 datasets = 300 total judge evaluations. The stratification ensures that the judge evaluation covers the full spectrum of memory difficulty rather than being dominated by the most frequent question type.

Three evaluation dimensions: Each sampled response is scored by Gemini 2.5 Flash on three dimensions using a 1–5 rubric.

Dimension	Definition	Score of 5	Score of 1
Contextual Coherence (1–5)	Does the response appropriately incorporate information from prior sessions retrieved by the memory backend?	The response seamlessly and accurately integrates retrieved memory.	The response ignores available memory entirely or contradicts it.
Answer Faithfulness (1–5)	Is every claim in the response grounded in either the retrieved memory records or the current query?	No hallucinated or ungrounded claims.	The response contains multiple claims with no basis in retrieved memory.
Conversational Appropriateness (1–5)	Is the response suitable in tone, format, and length for a personal assistant interaction?	The response reads naturally as a personal assistant reply.	The response is incoherent, excessively verbose, or inappropriate in register.

Table 3.6: LLM-as-Judge Evaluation Dimensions and Scoring Rubric

Bias mitigation: Known biases in LLM judges including verbosity bias, in which longer responses receive inflated scores, and position bias, in which the judge favors information presented earlier in the prompt are mitigated through two design choices. First, the judge prompt uses a structured rubric with explicit per-dimension criteria rather than open-ended comparative judgment, which Zheng et al. [30] demonstrate reduces verbosity bias. Second, retrieved memories and response text are presented in fixed positions across all evaluations, eliminating positional variation as a confound.

3.6 Statistical Analysis and Validation Framework

To confirm that observed performance differences across memory conditions are statistically significant rather than artifacts of random variation, the study employs a two-tier statistical framework: a repeated measures analysis for automated metrics and a descriptive-comparative analysis for LLM-as-Judge scores.

A one-way repeated measures ANOVA is employed to compare automated metric scores across three conditions: baseline (no memory), BM25 lexical retrieval, and dense semantic retrieval. This design is appropriate because the same model — Gemma 3 4B — is evaluated across all three conditions using identical question sets from LongMemEval and LoCoMo, satisfying the repeated measures requirement. The analysis is conducted separately for each dataset to preserve the interpretive distinction between LongMemEval’s controlled synthetic histories and LoCoMo’s real long-term conversational data. The null hypothesis for each metric states that mean performance scores are equal across all three memory conditions:

$$H_0 : \mu_{\text{baseline}} = \mu_{\text{BM25}} = \mu_{\text{dense}} \quad (3.4)$$

If the omnibus ANOVA yields a statistically significant result ($p < 0.05$), post-hoc pairwise comparisons are conducted using the Bonferroni correction to control the family-wise error rate. The Bonferroni-adjusted significance threshold for three pairwise comparisons is:

$$\alpha' = 0.05/3 = 0.0167 \quad (3.5)$$

The assumption of sphericity is assessed using Mauchly’s Test. If violated ($p < 0.05$), the Greenhouse-Geisser correction is applied to the degrees of freedom. This procedure is applied independently to each of the five automated metrics: Task Success Rate, Recall Accuracy, Memory Error Rate, Retrieval Latency, and Context Token Utilization.

3.7 Replicability

The detailed description of the experimental setup ensures that the study is replicable by professional peers in the domain of computational linguistics. The use of the SLM-bench framework ensures that future researchers are able to replicate the study using the same datasets (Section 3.4.1). Through this rigorous methodology, the research aims to bridge the gap between small-scale localized inference and high-fidelity long-term memory, enabling the next generation of personalized and privacy-centric artificial intelligence.

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